

When the trial can't *answer*.

Three cases where **clinical pharmacology** closed the gap.





When the clinical trial *can't answer.*

Three cases. Three trial limits. Three **clinical pharmacology** answers.

 UNTRIALABLE

From espresso to IV drip, **one question.**

How do we turn incomplete evidence into a defensible clinical pharmacology decision?



2004-2010

Dental Surgery (BDS) • JUST

Practicing clinician • Jordan License
PhD Scholarship
Patient-level intuition



2012-2015

PhD Clin Pharm • U. Minnesota

EHC modeling dissertation
ECP Fellowship
3 research awards



Summer 2014

Merck • QP2 • Intern

NLME simulation under uncertainty
Trial design inputs (sample size, dose range, endpoints)

Three cases. Three challenges. *One discipline at the center.*

Each case is a different drug and a different kind of challenge — pediatric, geographic, methodological. Quantitative pharmacology is what they share.

CASE 01

Ambrisentan

Pediatric PAH

EMA · PMDA · 2021

Exposure matching as the regulatory bridge when an efficacy trial cannot be run.

CASE 02

AI / ML

The Pharazi Foundation

PERSONAL RESEARCH · 2024 –

Architecture judgment for privacy-safe, audit-ready clinical pharmacology workflows.

CASE 03

Ivosidenib

India · CDSCO

RULE 101 WAIVER · 2025

A global Clin Pharm dossier can stand in for a local trial under reference-agency reliance.

CASE STUDY 01

Ambrisentan

Pediatric PAH · 39 patients ·
two agency approvals

When the pediatric trial is terminated — *can the pharmacokinetic bridge still support a pediatric label?*

A disrupted pediatric program tested whether exposure matching could still support a defensible dose.

ADULT DATA

380 adults across 6 studies — the mature adult PK anchor.

THE WHOLE STORY

HELD

Mar 2013

Juvenile rat brain-weight finding

REFRAMED

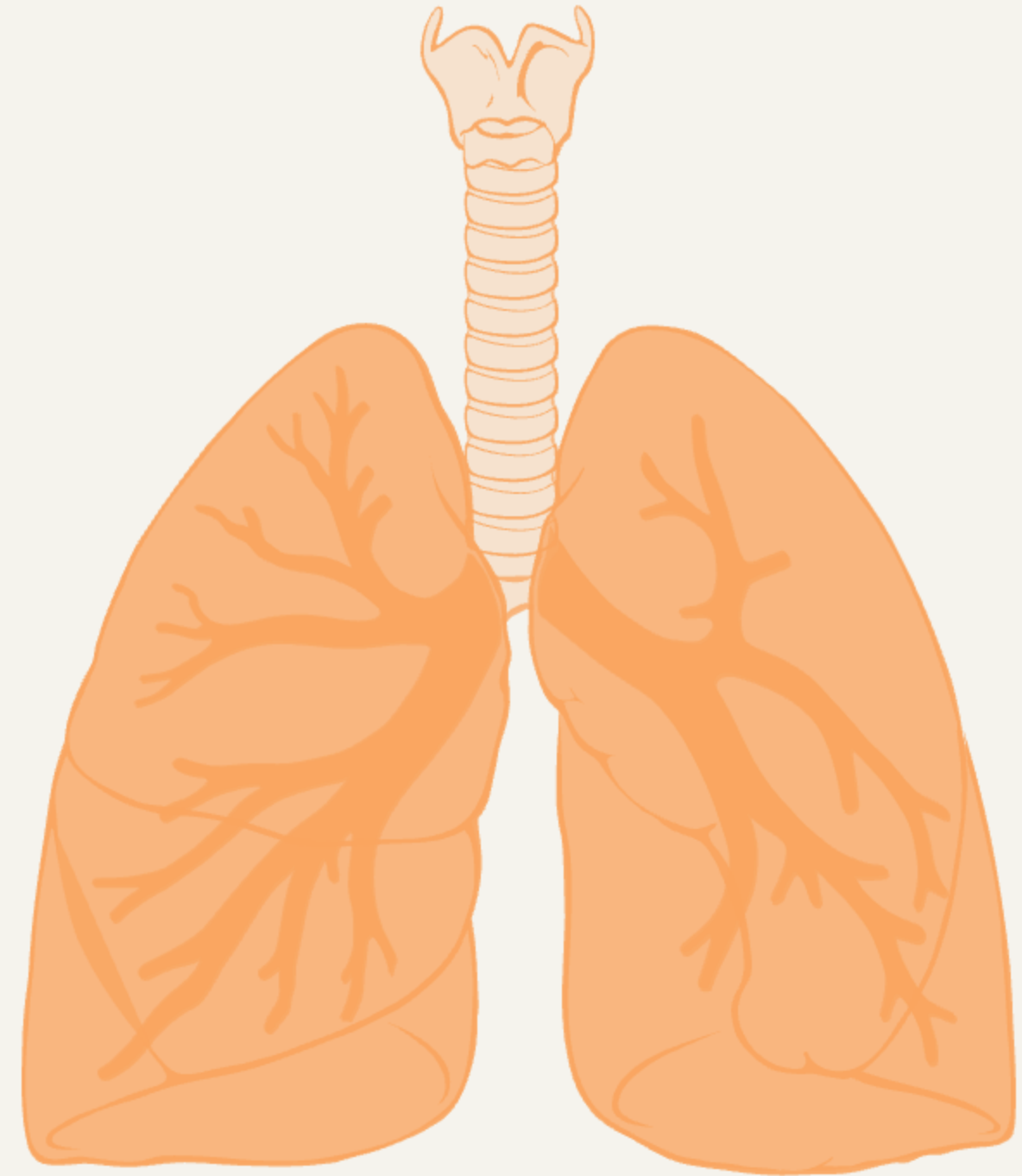
2014

STARTS-2 mortality signal

Pulmonary arterial hypertension is *small-vessel lung disease* that kills through right-heart failure.

The lumen narrows. PVR rises. The right ventricle hypertrophies, dilates, then fails.

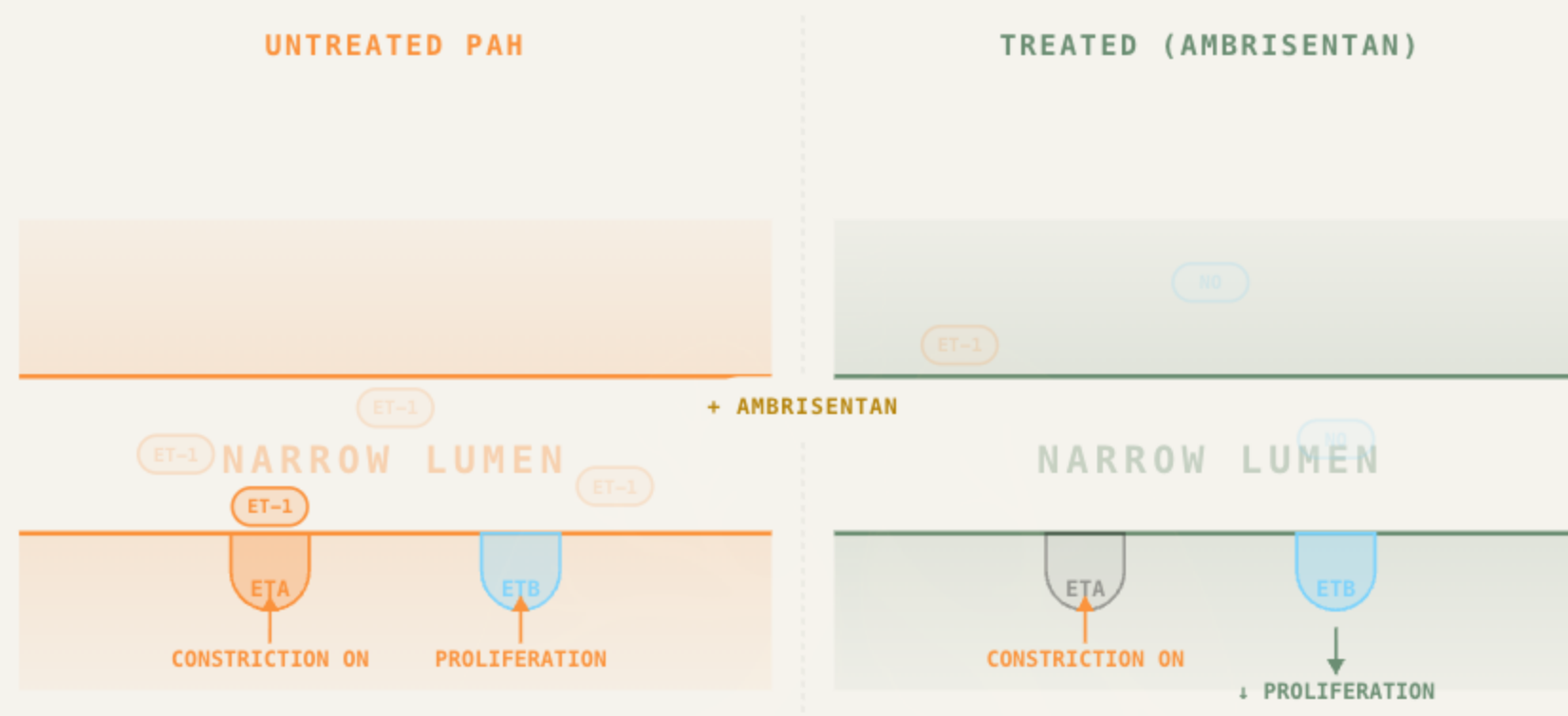
- 01 • The pathology **Vasoconstriction, smooth-muscle and endothelial proliferation, and in-situ thrombosis** — the lumen narrows.
- 02 • The hemodynamic $\text{mPAP} \geq 20 \text{ mmHg}$ · $\text{PVR} \geq 2 \text{ WU}$ · $\text{PAWP} \leq 15 \text{ mmHg}$ — pre-capillary, WHO Group 1.
- 03 • The pathways **Endothelin** ↑ · Nitric oxide ↓ · Prostacyclin ↓ · Activin / TGF-β dysregulated.
- 04 • The outcome Untreated median survival ≈ 2.8 years (*D'Alonzo 1991, NIH registry*).



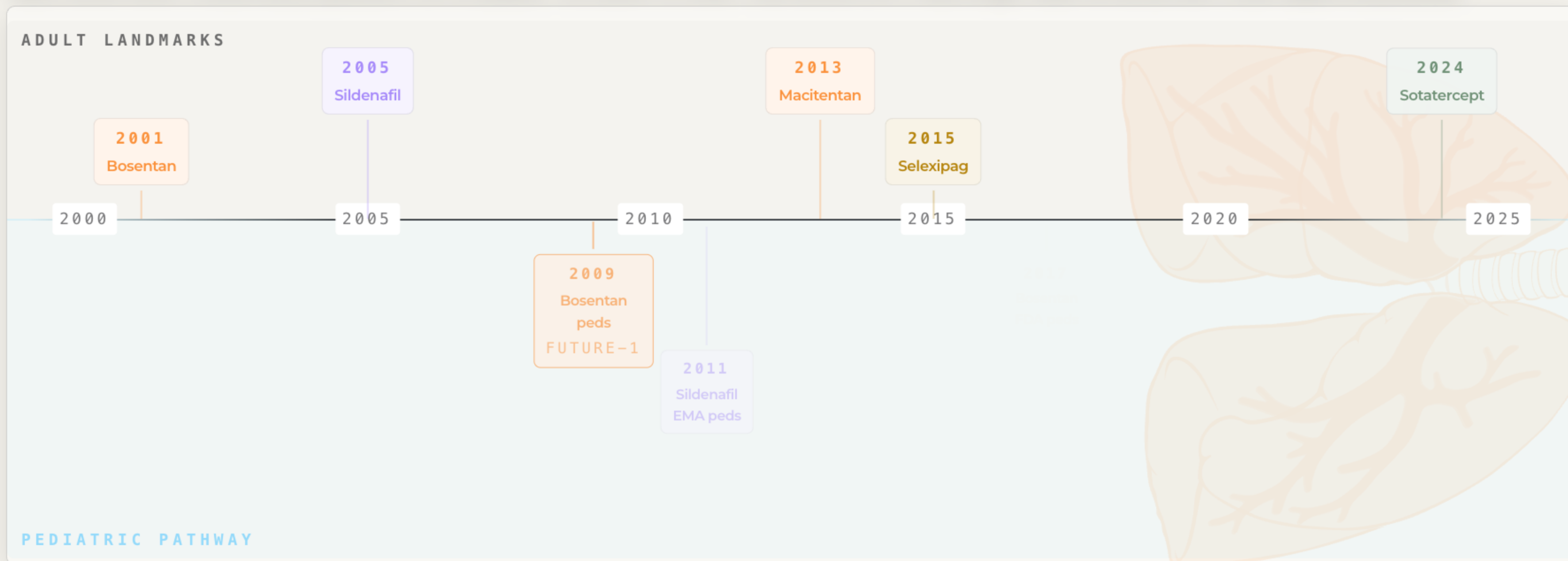
Ambrisentan blocks the **endothelin pathway** — the over-active vasoconstrictor and proliferative arm.

Three pathways drive PAH. *Ambrisentan selectively blocks the endothelin arm.*

Selectivity matters: ambrisentan blocks ETA (which drives vasoconstriction) but preserves ETB to allow vasodilation via NO.



Pediatric PAH moves slowly — *and ambrisentan's pediatric path took eight years.*



Adult evidence existed. *The pediatric efficacy-trial path did not.*

This is the pivot from "run the same trial" to "defend the pediatric dose quantitatively."

ADULT FOUNDATION • ANCHOR, NOT REPEAT		
ADULT ANCHOR	MECHANISM	DOSE TARGET
ARIES-1 / ARIES-2 established adult PAH efficacy	Selective ETA blockade; PAH biology is shared	Pediatric question becomes exposure matching, not a new efficacy trial

- 02 • POOLING

Heterogeneity

IPAH, CHD, CTD, and familial PAH in a small cohort.
- 03 • CONTROL ARM

Ethics

80% entered on baseline PAH therapy.
- 04 • ENDPOINT

6MWD transfer

Young children cannot perform it reliably; growth confounds change.
- 05 • PRECEDENT

Empirical record

STARTS-1: N=235; CPET peak VO2 primary p=0.056.

The clinical question wasn't *can we repeat ARIES in children?* The question was: *how do you defend a pediatric dose under these constraints?*

Three design decisions. Each one a *regulatory defense*.

A covariate screen designed to reject attractive but fragile explanations.

DECISION 01 • ANCHOR

Build on the adult model.



380 ADULTS

3,337 observations • structural anchor

39 PEDIATRIC

sparse sampling • refit, not rebuild

DECISION 02 • CONSTRAIN

ESTIMATED • N=39



exponents wobble • data-driven • unstable

FIXED • BIOLOGY-DRIVEN

Anderson-Holford 2008 • ICH E11A default

DECISION 03 • STAY PARSIMONIOUS

Test covariates — retain none.

12 TESTED

0 RETAINED

BILI	ALT	AST	ALP
GGT	CrCl	AGE	SEX
RACE	ETH	DOSE	TLAG

Extensive simulation, tight parsimony.

A 2-compartment structural model validated against 500 predictive replicates. Of 12 pre-specified covariates, only body weight scaled the exposure.

STRUCTURAL MODEL

2-CMT

ALLOMETRY • FIXED

$CL, Q \propto WT^{0.75}$ $Vc, Vp \propto WT^{1.0}$

70-KG REFERENCE • Anderson-Holford 2008 • ICH E11A 2025 default

CLICK TO ZOOM

PARAMETER ESTIMATES

CL/F, Vc/F

PARAMETER	ESTIMATE	UNIT	%RSE
CL/F	1.17	L/h	6.33
Vc/F	12.3	L	16.1
Q/F	0.457	L/h	21.1
Vp/F	81.3	L	24.5
Ka	2.46	1/h	25.7
tlag	0.525	h	14.7
Allom.	0.75 / 1.0	fixed	—

CLICK TO ZOOM

PCVPC

500 replicates

CLICK TO ZOOM

WORKFLOW

NONMEM

01 • BUILD

ADULT POPPK FOUNDATION

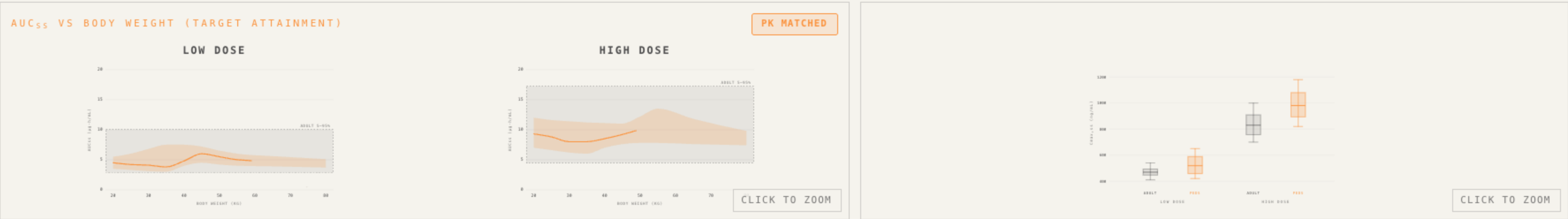
300 adults • 7 studies • 3,126 obs

CLICK TO ZOOM

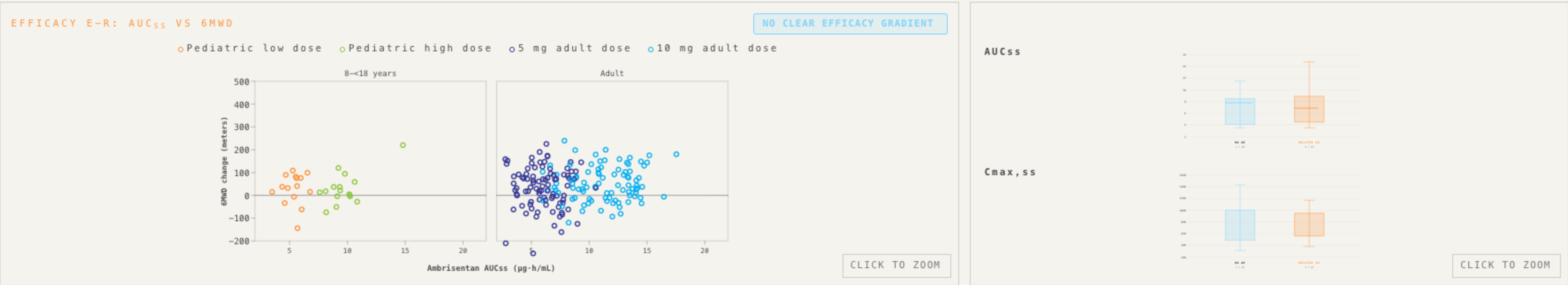
Exposure bridged; no clear E-R gradient.

Pediatric exposures land inside the adult target envelope, with no clear efficacy or safety gradient across observed exposure.

PK MATCHING



PK-PD CONTEXT



CONCLUSION: PK MATCHING HELD; NO CLEAR EXPOSURE-RESPONSE GRADIENT IN OBSERVED RANGE

AMB112529 survived three disruptions — *the dose bridge still held.*

Trial interruption, regulatory caution, and filing geography all pointed away from a conventional pediatric efficacy trial.

01 TRIAL

Enrollment stopped before efficacy could close.

HOLD 2013–2017 • TERMINATION 2019 • 39 / 66 EVALUABLE

Pediatric dataset became confirmatory.

SO WHAT

Dose rationale: **adult anchor + pediatric PK.**

02 REGULATORY

Pediatric dosing was under caution.

REVIEW 2017–2021

Empirical escalation was hard to defend.

SO WHAT

Exposure matching was the **cleaner lane.**

03 FILING

Label outcomes followed filing geography.

EMA + PMDA PROCEEDED

Submissions diverged; labels diverged.

SO WHAT

Public outcome: **EMA + PMDA pediatric approvals.**

The dose was defended by *exposure matching, not a repeat efficacy trial.*

Adult anchor → prespecified PopPK bridge → pediatric exposure confirmation.

01 • ADULT ANCHOR

380 adults

6 studies
3,126 PK observations

02 • POPPK BRIDGE

2-cmt oral

absorption + lag
 $CL \propto WT^{0.75} \cdot V \propto WT^{1.0}$

03 • SIMULATION

AUC by weight

Predict pediatric exposure
Target: adult AUCss range

04 • TRIAL PK

39 children

211 sparse PK observations
ages 8 to <18 years

05 • EXPOSURE MATCH

-3% low dose

+0.3% high dose
within adult AUCss range



EMA and PMDA approved pediatric ambrisentan in 2021. *ICH E11A codified the framework in 2024.*

The framework the case demonstrated has since become the codified standard for pediatric extrapolation.

PIN 01

EMA 2021

Pediatric Volibris · 8–17 years · 3 weight bands × 2 dose levels (2.5–10 mg QD)

PIN 02

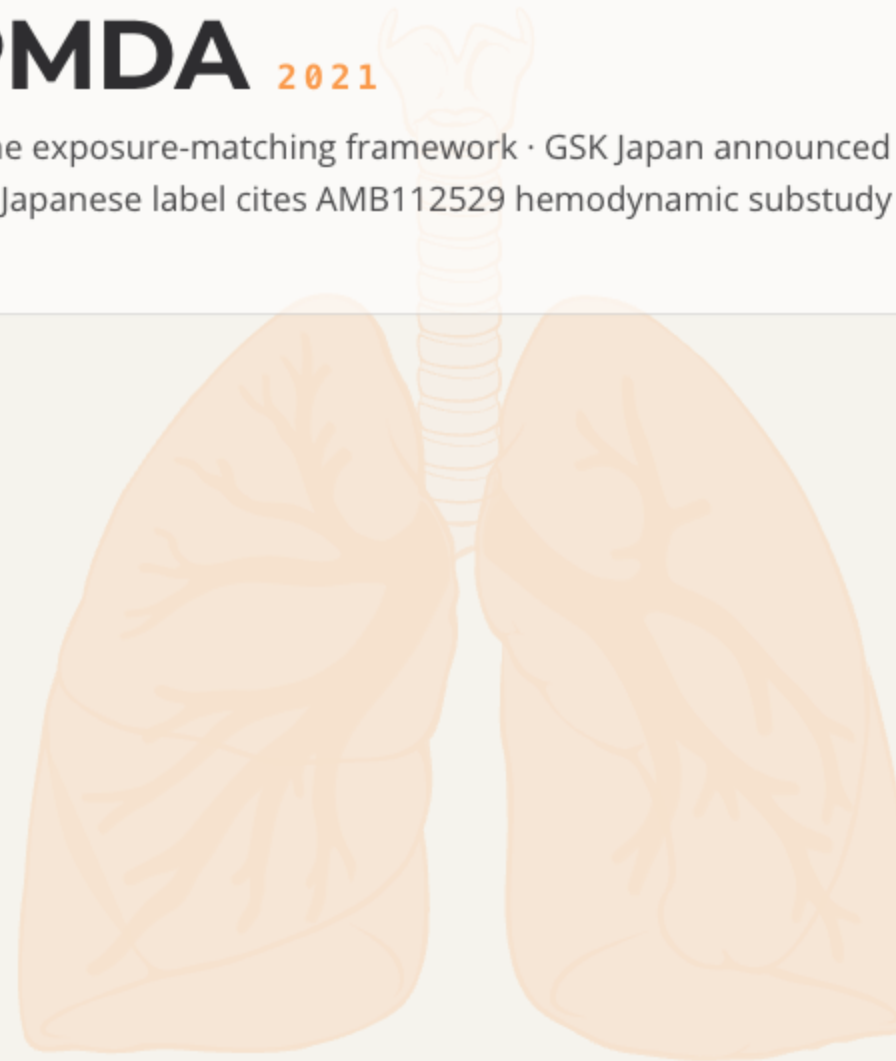
PMDA 2021

Same exposure-matching framework · GSK Japan announced Mar 23 · Japanese label cites AMB112529 hemodynamic substudy

PIN 03

ICH E11A December 2024

Extrapolation continuum codified · where similarity is high → exposure matching can carry more of the inference



When the trial cannot deliver the dose, *the quantitative bridge makes it defensible.*

What this case teaches — beyond ambrisentan, beyond pediatric PAH.

CASE STUDY 02

AI the Pharazi

Pharazi.ai · open-source
project

The regulatory floor is *set*

Three guidances. Three jurisdictions. An eighteen-month enforcement window.

FDA • DRAFT GUIDANCE

6 JAN 2025

4 STEP RISK FRAMEWORK

AI IN REGULATORY SUBMISSIONS

Risk-based credibility framework establishing how AI/ML evidence is reviewed.

FDA • EMA JOINT PRINCIPLES

12 JAN 2026

0 GUIDING PRINCIPLES

CROSS-JURISDICTIONAL • BINDING

Underpins every future AI guidance in the U.S. and EU.

ICH • M15

16 JAN 2026

EU EFFECTIVE 23 JUL 2026

AI/ML covered alongside PopPK, PBPK, and QSP as legitimate MIDD methodologies.

Each system owns *one or two* stages

The market is moving above the regulatory floor — fragmented, manual, no end-to-end foundation.

Apollo-AI

Pfizer + InsightRX • Shahin et al. CTS 2025

Quantitative clinical pharmacology workflows

MIDD STAGES

3 / 7

PharmaSwarm

arXiv · 2025

Hypothesis generation

MIDD STAGES

1/7

Pumas Suite

Pumas-AI · commercial

Pumas-bound add-ons (DeepPumas, AskPumas, PumasAide

MIDD STAGES

2 / 7

Prompt-to-Pill

Vichentijevikj et al. · 2026

Discovery + virtual trial

MIDD STAGES

2 /

pyDarwin

Open source · Certar

PopPK structural search

MIDD STAGES

1/3

PEARL

Buffalo group - 2026

Regulatory RAG only

MIDD STAGES

1/7

QSP-Copilot

Discovery has multi-agent.
QCP has Apollo-AI.

MOVEMENT II • ARCHITECTURE

The foundation *exists.*

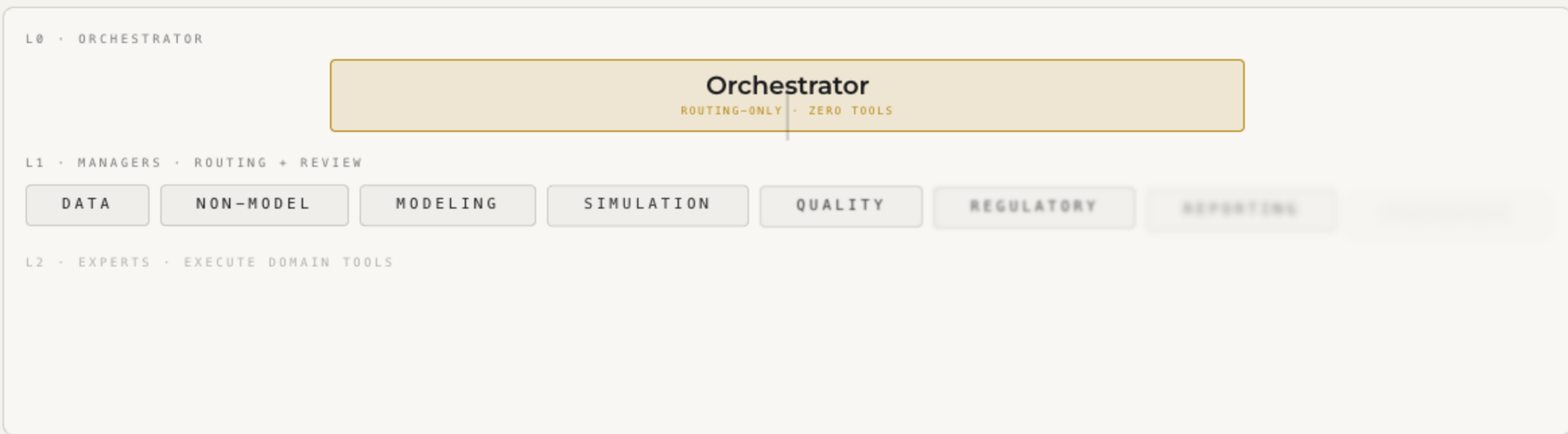
DESIGN PRINCIPLES • WORKING SYSTEM • SCALABILITY

MOVEMENT II BEGINS

M1 • THE VISION • 4M M2 • THE ARCHITECTURE • 18M M3 • THE FUTURE • 8M

Centralized Hierarchy

Independent agents amplify errors 17× as scope grows; centralized hierarchies contain that to 4×.



Structural Privacy

Privacy is a property of the code, not a runtime policy. Raw data has no callable path to the LLM.

What the model sees

Schema Extractor - zero LLM calls

COMPUTATION ENGINE · LOCAL

Where the data lives

Cryptographic Audit

***Every state mutation, every routing decision,
every approval — bound into one verifiable hash
chain.***



Versioned Workflow + Coordination Discipline

*Workflows are hash-anchored execution plans;
coordination cost is bounded by deliberate
constraints.*

EXAMPLE • STANDARD POPPK SOP • V2.1

11 STEPS



We don't scale by rebuilding. We scale by *registering*.

Same hierarchy. Same privacy firewall. Same audit chain. Additive only — across every domain.

- PHASE 1
- PHASE 2
- PHASE 3

← / SPACE → TO ADVANCE

PHASE 1 · CLINICAL PHARMACOLOGY + PHARMACOMETRICS

- NCA
- POPPK
- PKPD
- E-R
- QC

SHARED INFRASTRUCTURE · BUILT ONCE

- Schema Extractor
- Audit Chain
- QC Debate
- Manager Review
- Reg RAG
- Report Gen
- HITL Gate
- State Bus

MOVEMENT 3 OF 3 · 8 MINUTES

The Future

Five components. Two synthesis views. The publication close.

ENTERING MOVEMENT 3 · THE FUTURE

M1 · THE VISION · 4M

M2 · THE ARCHITECTURE · 18M

M3 · THE FUTURE · 8M

It runs. The chat triggers the chain.

One sentence in. The orchestration log streams routing decisions in real time.

USER INPUT · CHAT

user@org · 14:02 · "Run NCA on my dataset, 0-24h, linear-up/log-down. Stratify by formulation."

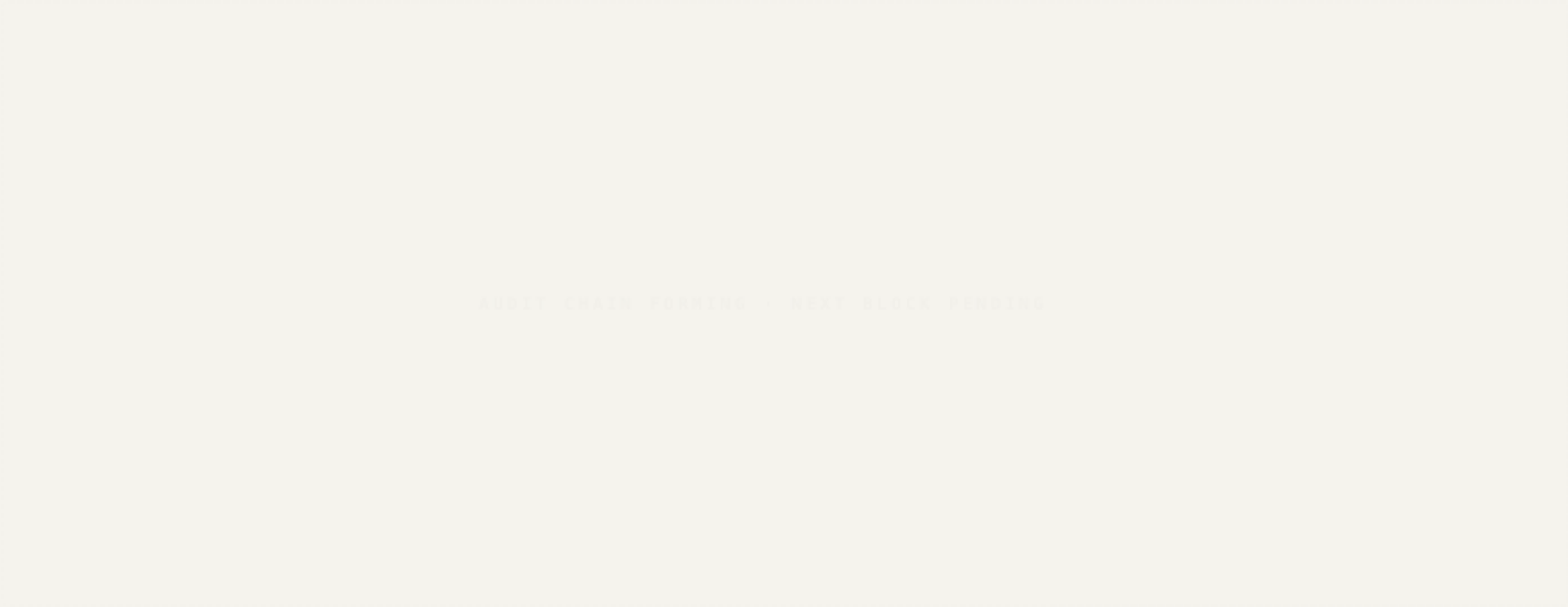
ORCHESTRATION LOG · LIVE

```
→ Routing: non_model_manager
→ Delegating: nca_expert
```

NCA PARAMETERS · COMPUTING...



AUDIT CHAIN · ANCHORING...



It runs. The chain anchors. The numbers ship.

Vercel · Railway · Supabase · production deployment · April 2026

USER INPUT · CHAT

user@org · 14:02 · "Run NCA on my dataset, 0-24h, linear-up/log-down. Stratify by formulation."

ORCHESTRATION LOG · COMPLETE

→ Complexity: MODERATE

→ Routing: non_model_manager

→ Delegating: nca_expert

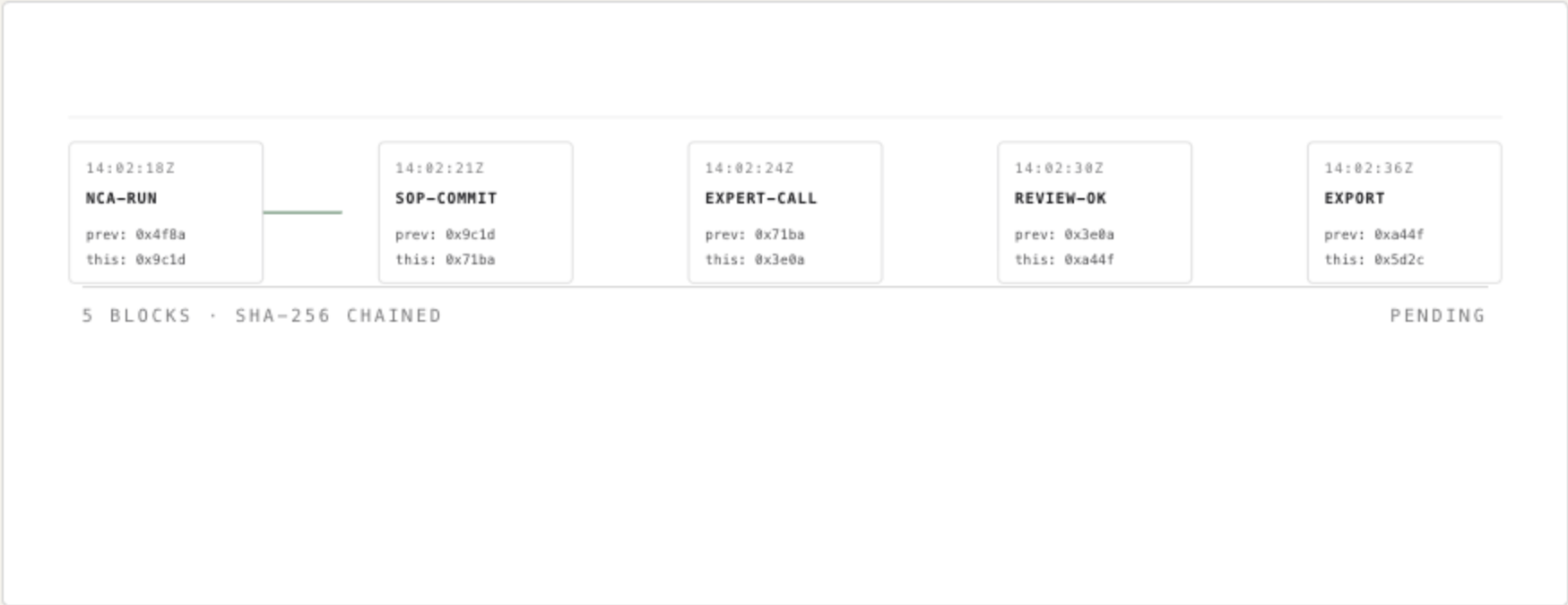
→ Tools: calculate_nca, plot_pk_profile, compute_t_half

→ QC review: PASS · 3/3 vote

→ Audit chain: anchored to sop.nca_v1_2

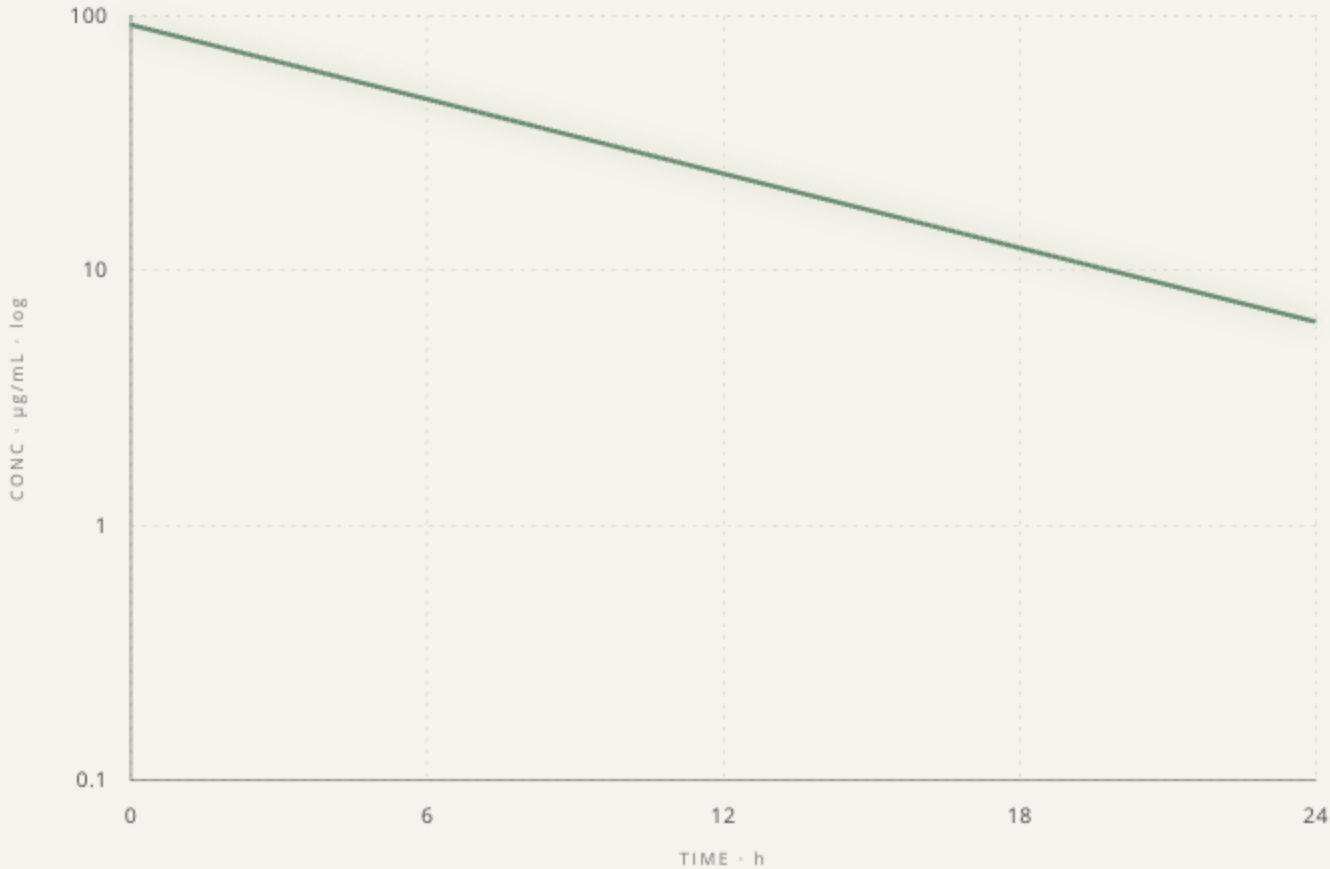
DEPLOYED · LIVE

AUDIT CHAIN · VERIFY SWEEP



Non-Compartmental *Analysis.*

Concentration in. Parameters out. Stamped, hashed, reproducible.

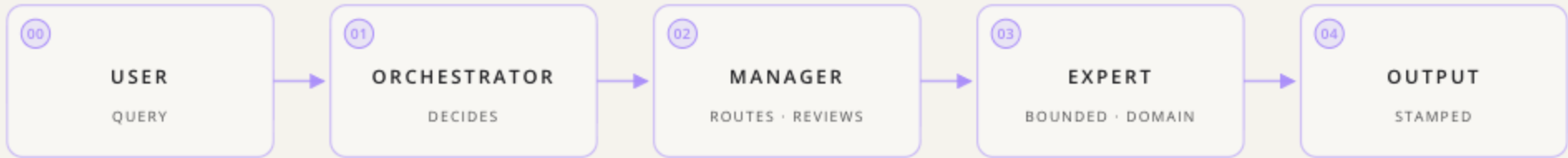


● NOW · DRAWING CURVE

● NCA CLINICAL · PRIMITIVE	01 / 05
Data Flow ORCHESTRATION · TOPOLOGY	02 / 05
Audit Chain INTEGRITY · INFRASTRUCTURE	03 / 05
Privacy Wall PRIVACY · INFRASTRUCTURE	04 / 05
Marketplace SOP OPERATIONS · LIFECYCLE	05 / 05

Data *Flow*.

Every query routes top-down. No expert acts unbounded.



TOKEN CACHE 0%

hot context · 14 keys

AUDIT LOG 0%

hash-chain · sha-256

OUTPUT 0%

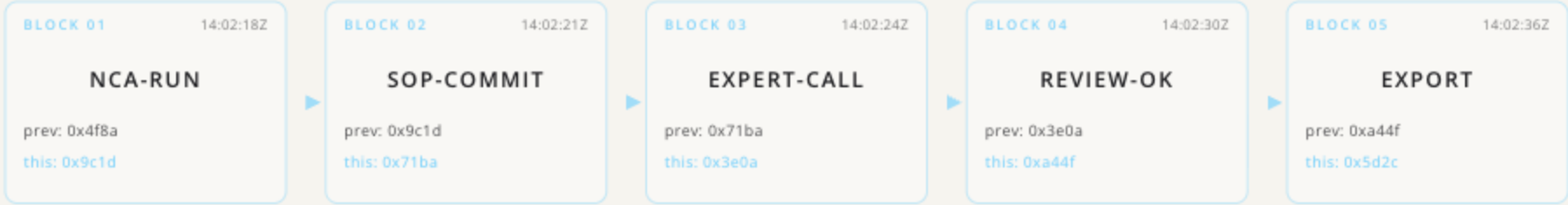
stamped · v1.0

Data Flow ORCHESTRATION · TOPOLOGY 02 / 05
Audit Chain INTEGRITY · INFRASTRUCTURE 03 / 05
Privacy Wall PRIVACY · INFRASTRUCTURE 04 / 05
Marketplace SOP OPERATIONS · LIFECYCLE 05 / 05

Audit Chain.

Every decision binds into one hash chain. Tampering is observable.

4 NOW · BINDING ENTRIES



SUBSTRATE · 5 COMPONENTS

NCA CLINICAL · PRIMITIVE	01 / 05
Data Flow ORCHESTRATION · TOPOLOGY	02 / 05
Audit Chain INTEGRITY · INFRASTRUCTURE	03 / 05
Privacy Wall PRIVACY · INFRASTRUCTURE	04 / 05
Marketplace SOP OPERATIONS · LIFECYCLE	05 / 05

SAME HIERARCHY · SAME PRIVACY · SAME AUDIT

PopPK inherits the *same foundation*.

Covariate strategy → simulation → diagnostics → audit. Zero new infrastructure.

COVARIATE STRATEGY · INCLUSION MATRIX

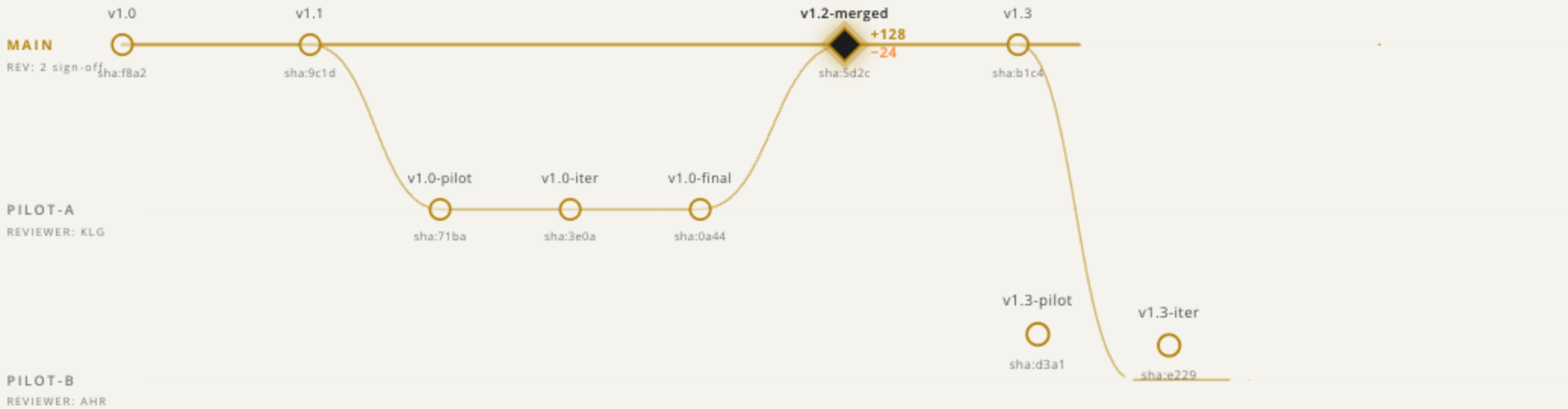
DOSE-EXPOSURE WATERFALL · N=200 SIMULATED

AUDIT TRAIL - ANCHORED TO SOP.POPPK_V2_3

Marketplace *SOP*.

Versioned execution plans. Git commits for analytical decisions.

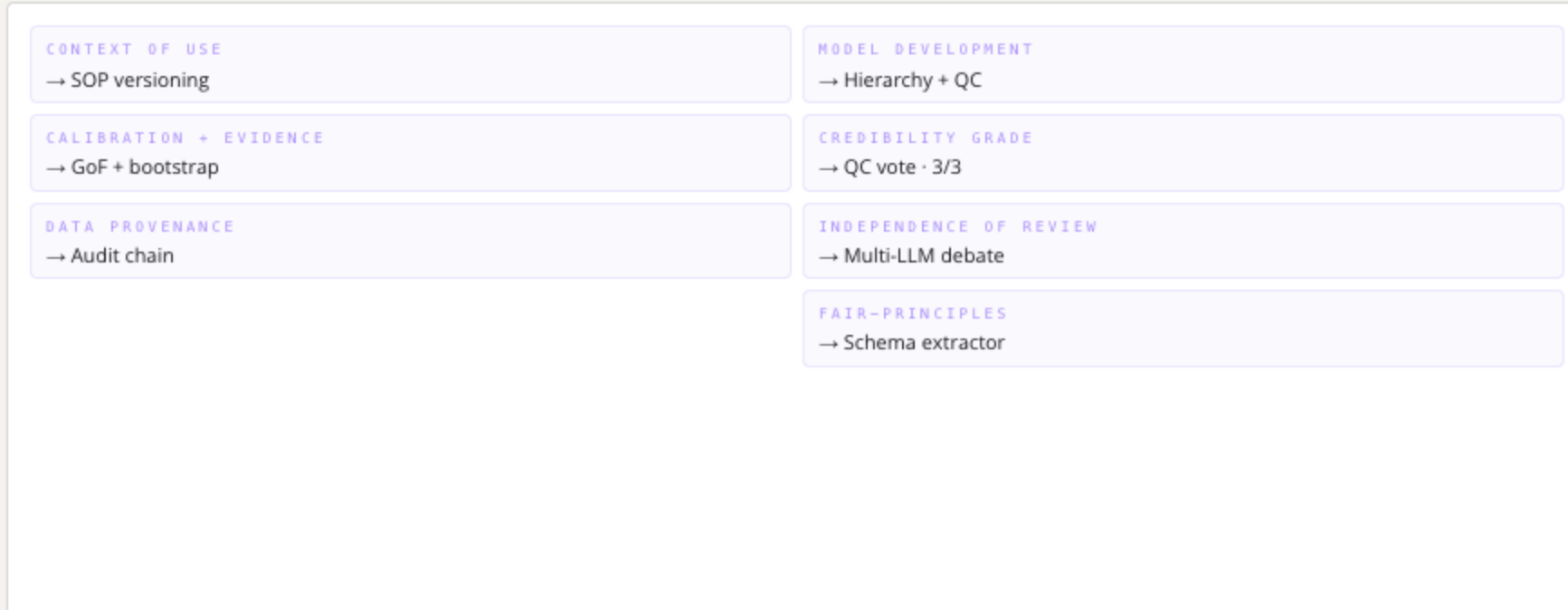
NOW · MAIN · V1.3



NCA CLINICAL · PRIMITIVE	01 / 05
Data Flow ORCHESTRATION · TOPOLOGY	02 / 05
Audit Chain INTEGRITY · INFRASTRUCTURE	03 / 05
Privacy Wall PRIVACY · INFRASTRUCTURE	04 / 05
Marketplace SOP OPERATIONS · LIFECYCLE	05 / 05

The same chain anchors *regulatory text*.

ICH M15 expectations · CTD modules · agency engagement — all derive from the foundation.



One sample. One number. One regulatory paragraph. *One chain.*

Provenance is structural, not procedural. Every claim resolves to a verifiable root hash.

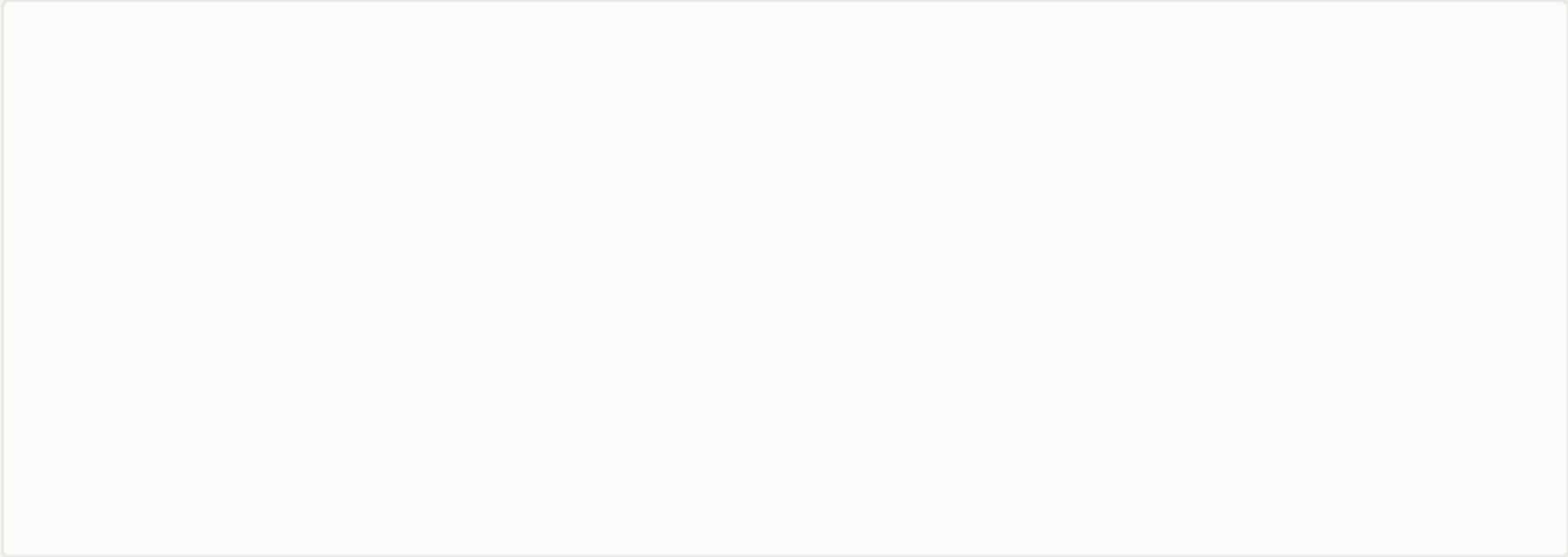
PROVENANCE TREE · 4 TIERS



TRACEABILITY MATRIX · CLAIMS × BLOCKS

CLAIM	L1	L2	L3	L4
COVARIATE INCLUSION	-	✓	✓	-
EXPOSURE TARGET HIT	-	-	✓	✓

REGRESSION + INSPECTOR ALERTS



The foundation, *composed.*

One router. One firewall. One chain. One registry. Five components — all anchored to the same root.

COMPONENTS · 01 / 05

DETAIL · NON-MODEL MANAGER · ROUTER

ANCHOR · ROOT HASH

- 01

Centralized Hierarchy

NON-MODEL MANAGER · ROUTER
- 02

Privacy Wall

STRUCTURAL FIREWALL · SCHEMA-ONLY
- 03

NCA Computation

DETERMINISTIC TOOL · LINEAR-UP/LOG-DOWN
- 04

Audit Chain

CRYPTOGRAPHIC LINEAGE · SHA-256
- 05

Marketplace SOP

VERSIONED PLANS · SOP REGISTRY

01 Centralized Hierarchy

ROUTING DECISION

3.2s

AVG

A single router examines complexity, intent, and data shape. It hands off to the correct domain expert (NCA, PopPK, PKPD). No agent talks directly to another agent. This is how the chain stays auditable.

WITHOUT FOUNDATION

Free-form A2A · routing drift · no QC handoff vote

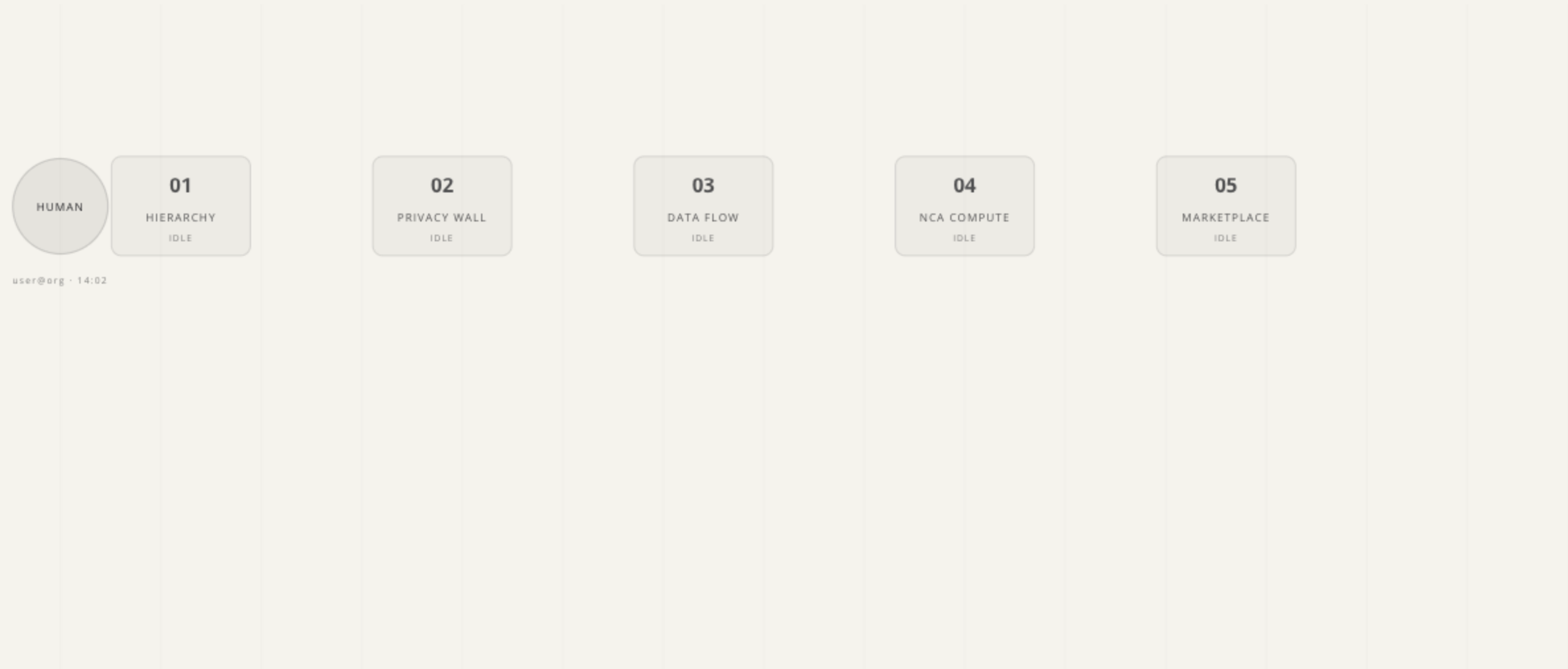
WITH PHARAZI

Single router · structural · 3-of-3 vote at every gate

ROOT	0x4f3a8e91
LINKS	5 / 5 verified
DRIFT	0 detected
M15	aligned

One request becomes *a replayable evidence trail.*

Five components · five principles · one chain. Time bar at the bottom narrates each beat.



Live at *pharazi.ai*

The reference architecture, the working system, the ecosystem — open and addressable.

• MANUSCRIPT · CPT:PSP · IN PREP · APRIL 2026

Regulatory framework? *Exists.*

Architectural foundation? *Built.*

Working system? *Deployed.*

The foundation *holds.* The next chapter *scales it.*

• Manuscript · CPT:PSP · IN PREP

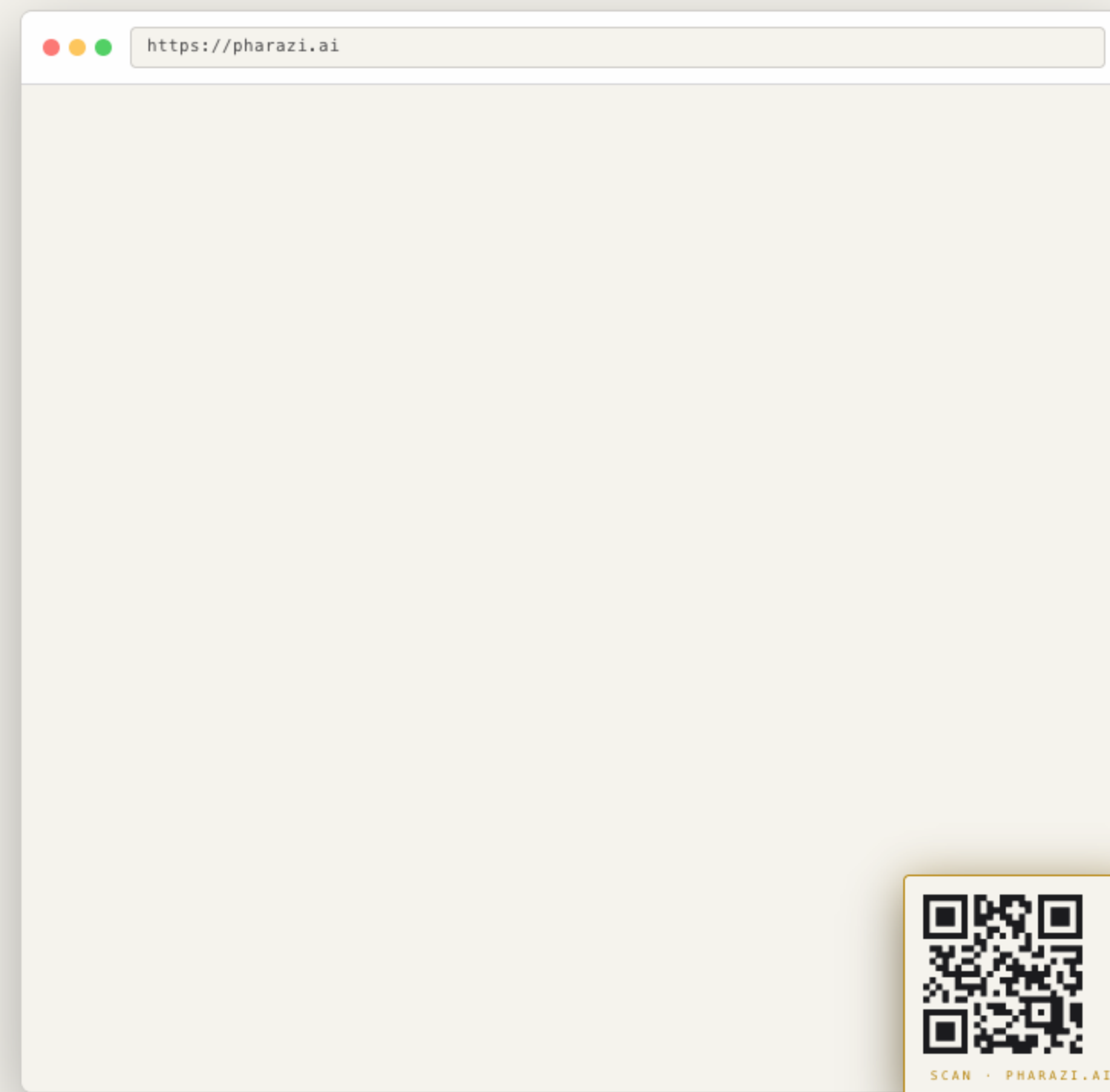
• github.com/pharazi · LIVE

• pharazi.ai · FRAMEWORK HOME

• clinpharm.ai · COMMUNITY

• R + Python pkgs · ROADMAP

LIVE · PHARAZI.AI



CASE STUDY 03

Ivosidenib

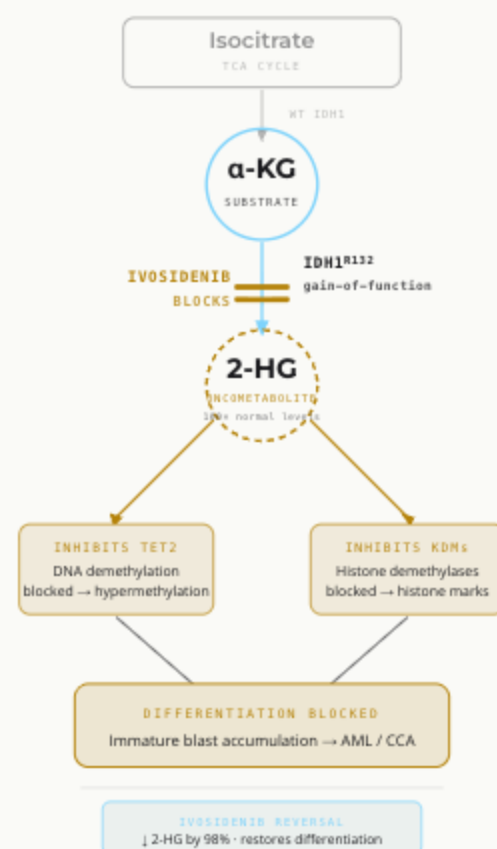
Regulatory strategy in India

42 countries said yes. **India** asked: *prove it locally*.



IDH1-mutant cancers: small populations, *no targeted option before ivosidenib.*

MECHANISM — MUTANT IDH1 PATHWAY



6–10%

ACUTE MYELOID LEUKEMIA

IDH1-mutated prevalence across AML subtypes

~20 K new AML/yr US • 1,200–2,000 IDH1+

~13%

CHOLANGIOCARCINOMA

of intrahepatic CCA harbor IDH1 mutations

~8 K new CCA/yr US • ~1,000 IDH1+

R132 HOTSPOT

~95%+

of IDH1 mutations at codon R132. One mutation, one drug target.

SOMATIC

Acquired

in tumor, not inherited. Disease defined by the mutation.

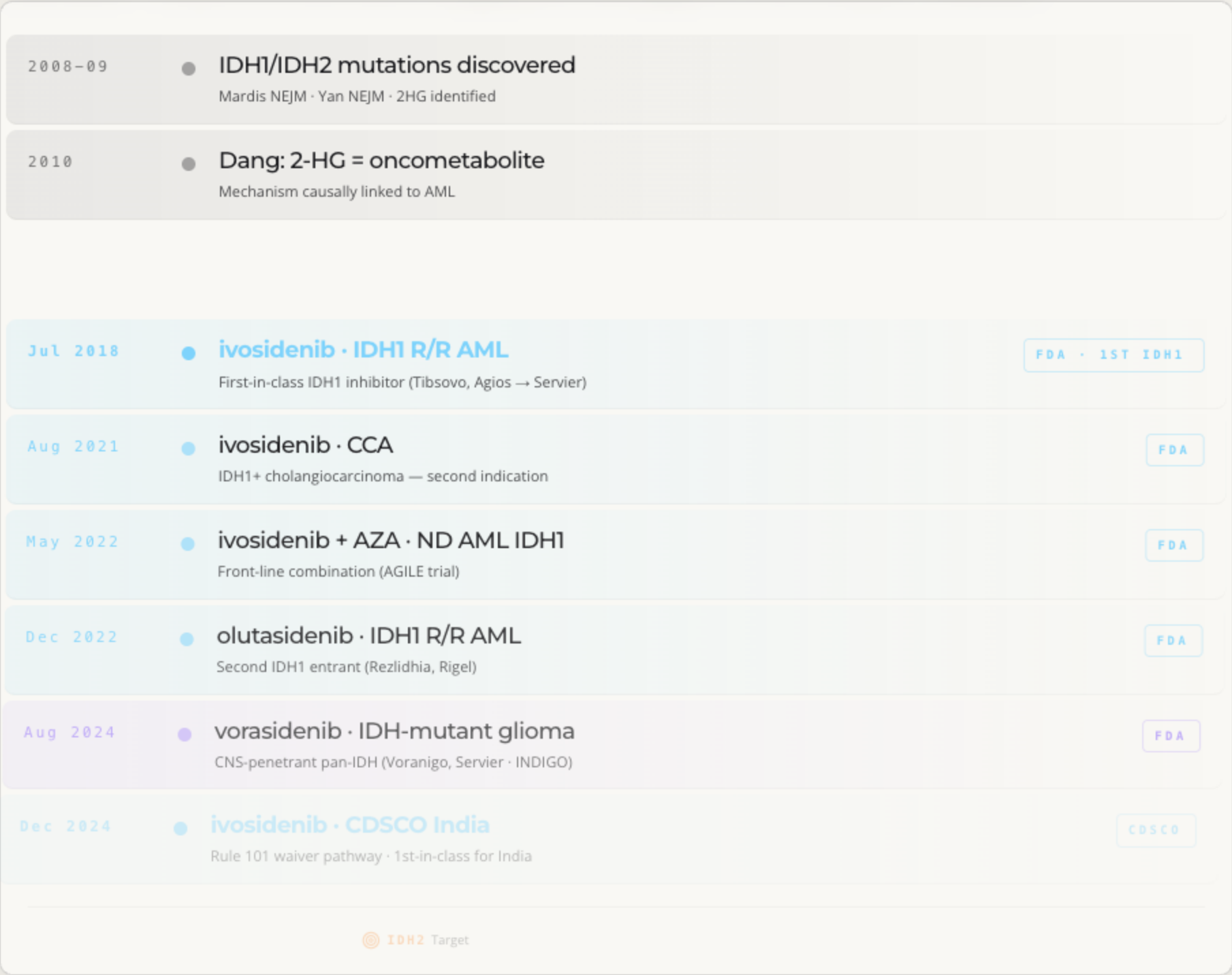
R/R AML • PRE-2018

3–9 mo

Median OS. No targeted option.

Sixteen years from discovery to *first-in-class in India*.

IDH1/IDH2 mutations were named in 2008. The first FDA approval came nine years later. The CDSCO Rule-101 approval — sixteen years after the gene was



IDH2

enasidenib

Bristol-Myers Squibb · AML

India's regulatory path got *shorter* on 7 August 2024.

Rule 101 created a Clin-Pharm-led waiver pathway. Ivosidenib qualified.

START

Indian MAA filed

December 2024 — the SEC asked for *local PK/PD*.

The science to answer existed. The science to comply did not.

The mutation is *somatic*. The drug *binds* it. The cascade *reverses*.

Three claims, one canvas — and four IDH-inhibitor competitors for context.

WHERE THE MUTATION LIVES

Somatic IDH1 R132 · tumor tissue only · not in germline DNA



WT



TUMOR



WT

If germline → all 3 cells marked → drug response would track ancestry. *Not this case.*

IDH-INHIBITOR LANDSCAPE

Public approvals · FDA Drugs@FDA verified 2026

Ivosidenib	IDH1
AML · CCA · FDA 2018 · <i>India 2025 — this case</i>	
Enasidenib	IDH2
AML R/R · FDA 2017 · <i>first-in-class IDH</i>	
Olutasidenib	IDH1
AML R/R · FDA 2022 · <i>2nd IDH1 entrant</i>	
Vorasidenib	IDH1/2
Grade 2 glioma · FDA Aug 2024 · <i>brain-penetrant</i>	

INSIDE THE TUMOR CELL — CASCADE FLIPS ON DRUG BINDING

Disease state → ivosidenib intervenes → drug state · *differentiation resumes*

Mechanism · Dang 2009 · TET2/JmjC inhibition · Figueroa 2010 · α -KG displacement

Six lines converge — across *PK, PD, intrinsic, extrinsic, regulatory, and mechanism.*

The ICH E5(R1) framework, fully populated.

PILLAR 01 MOA

**IDH1
R132**
somatic mutation



Drug binds mutant enzyme directly. *Biology conserved.*

Dang
Cancer Cell 2009

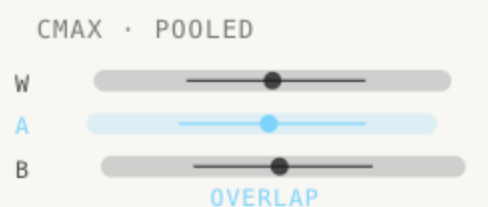
PILLAR 02 PK SIMILARITY

n = 253

race not significant

Pooled phase 1 + AGILE PK data, n=253. Linear PK confirmed across ethnic groups.

Jiang et al.
CTS 2021



PILLAR 03 ER SIMILARITY

Flat

across range

No exposure-AE or exposure-efficacy relationship. Wide TI • 500 mg QD covers range.

Phase 1 + AGILE pivotal



PILLAR 04 INTRINSIC

No impact

organ fn • age • sex

No demographic dose adjustment. CYP polymorphism characterized in DDI program.

Tibsovo USPI • EMA EPAR

✓ Renal

✓ Hepatic

✓ Age

✓ Sex

PILLAR 05 EXTRINSIC

No impact

food • DDIs • comeds

Extrinsic factors were characterized and managed; none required India-specific dose changes.

Tibsovo USPI
EMA EPAR

✓ FOOD

✓ DDI

✓ COMED

✓ LIFE

CHARACTERIZED • MANAGED

PILLAR 06 GLOBAL REG

30+

jurisdictions • 8 yrs

Multi-agency PV • 8 yrs surveillance • no ethnicity-specific signals.

FDA Orange Book
EMA EPAR • multi-agency



Public record, then the reversal — 14 May 2025.

FDA labels, EMA EPAR, Rule 101, and India authorization in one timeline.

INDIA AUTHORIZATION · PUBLIC RECORD



CDSCO MARKETING AUTHORIZATION

PUBLIC REGULATORY TIMELINE · MERGED FROM PRIOR VELOCITY SLIDE

What we *shipped*, and what we did *not*.

Naming the gap is the credibility move — not the disqualifying one.

WHAT WE SHIPPED

✓ 4

Quantitative pharmacology built the bridge — partners carried it through.

The waiver worked because evidence ownership and execution ownership were separated clearly.

QUANTITATIVE PHARMACOLOGY

- Translate the ICH E5 question into testable evidence
- Mechanism-first ethnic-sensitivity rationale
- Integrate PopPK · E-R · PBPK · intrinsic/extrinsic factors
- Define Phase 4 PK/PD as the residual-uncertainty plan

PARTNER FUNCTIONS

- Regulatory affairs**
Agency pathway, filing mechanics, and formal responses
- Medical / PV**
Post-marketing surveillance and local follow-up plan
- Regulatory writing**
Submission narrative, response package, and traceable record
- India affiliate**
In-country execution, meeting logistics, and access launch plan

The science *was the bridge.*

What this case teaches — beyond ivosidenib, beyond India.

- 01 — *Mechanism* is the foundation.
- 02 — *Convergence* is the case.
- 03 — *Transparency* earns trust.

◆ Wherever local trials aren't feasible — the *Clin Pharm dossier* becomes the bridge.

When the trial isn't the answer, *the framework is.*

Three cases, three obstacles, one discipline — quantitative evidence when the trial alone cannot carry the answer.

CASE 01 · AMBRISENTAN

Trial untrialable.

Adult anchor + pediatric PopPK + exposure match.

EMA + PMDA · 2021 · pediatric PAH label

CASE 02 · IVOSIDENIB

Trial unavailable.

Six-pillar regulatory dossier — MOA-anchored.

CDSCO · 14 May 2025 · India waiver granted

CASE 03 · PHARMAGENT

Workflow unbuilt.

13 agents · 151 tools · privacy + audit by construction.

Replayable evidence trail · M15-aligned documentation

Three validated *foundations*.

A track record of taking complex methodologies and turning them into defensible, regulatory-grade evidence.



PAH • PEDIATRICS

Pediatric extrapolation.

A regulator-tested framework for defining adult anchors, pediatric PK, and defensible dose logic.



ONCOLOGY • BRIDGING

Global-to-local bridging.

Translating disease mechanism, PK/PD, and covariates into regional reliance dossiers when local trials are infeasible.



AI INFRASTRUCTURE

Audit-backed architecture.

Designing scalable computational tools around immutable ledgers and strict human-in-the-loop workflows.

I would be glad to take your questions.

Three cases, one discipline — I'm here to discuss any of them in more depth, or to talk about how that translates to QP2 / CMD priorities.

Thank you.

*“The trial is not the only
answer.”*

— BACKUP · CASE STUDY 01 · AMBRISENTAN

Ambrisentan

Defense library · pull on demand

— BACKUP TYPE 1

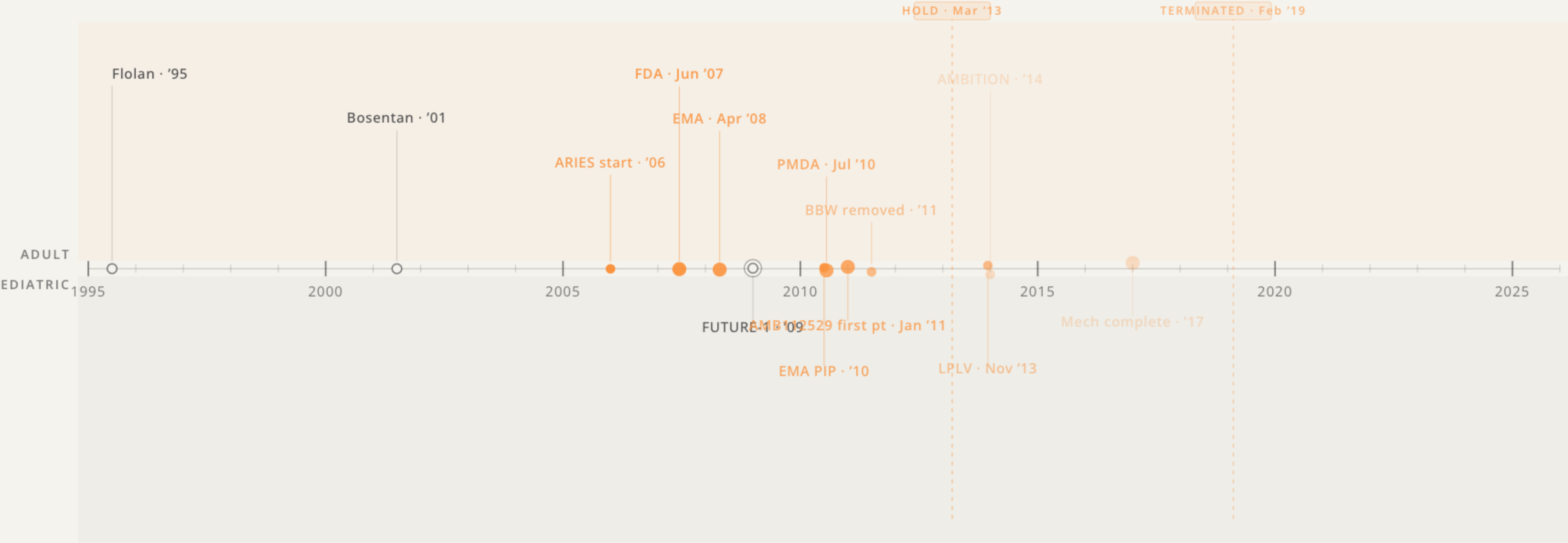
Historical Context

Why the PAH program had to evolve the way it did, and how ambrisentan's pediatric arc fits the broader timeline of disease, label, and regulatory framework precedents.

HOSTILE PROBE *"Didn't FUTURE-1 already do this in 2009? Why was the AMB12529 design needed at all?"*

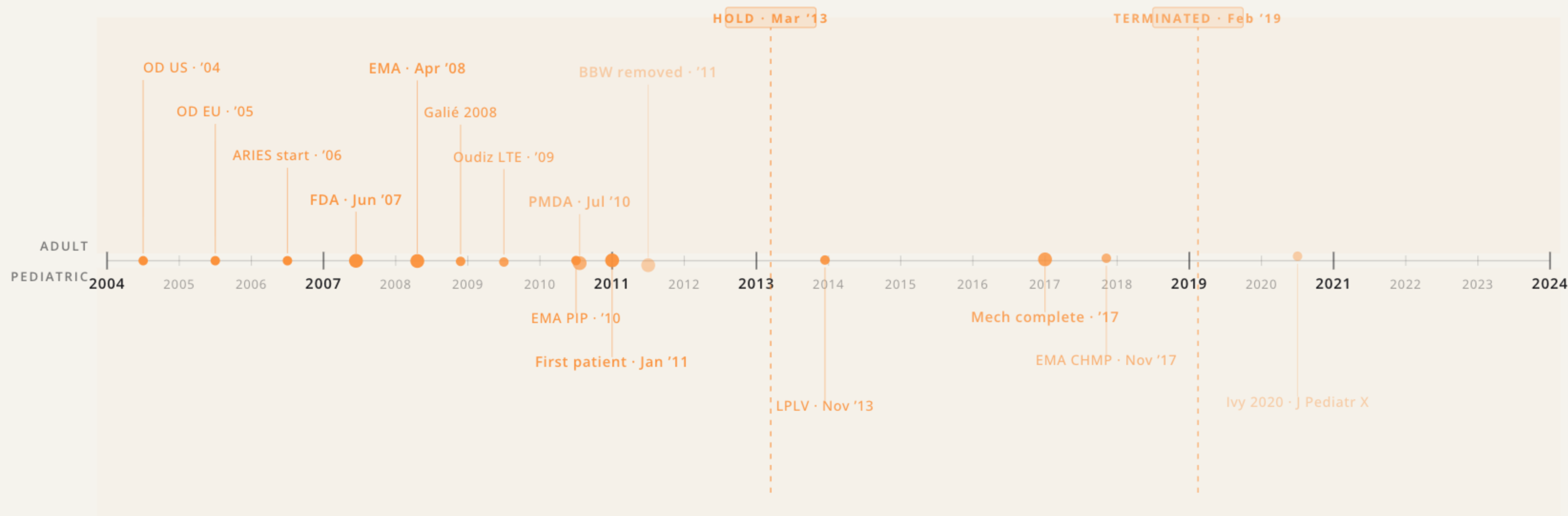
Ambrisentan in PAH · *adult and pediatric tracks, 1995–2026.*

Adult lifecycle reads as a linear approval cascade. Pediatric track absorbs two disruptions and resolves into a framework — codified four years later by ICH E11A.



Ambrisentan only · *twelve adult years on top*, one pediatric trial below.

Adult: orphan status → ARIES → three regional approvals → label update → lifecycle. Pediatric: one trial, held, then terminated, then resolved into dual approval. PopPK paper formalises the method afterward.



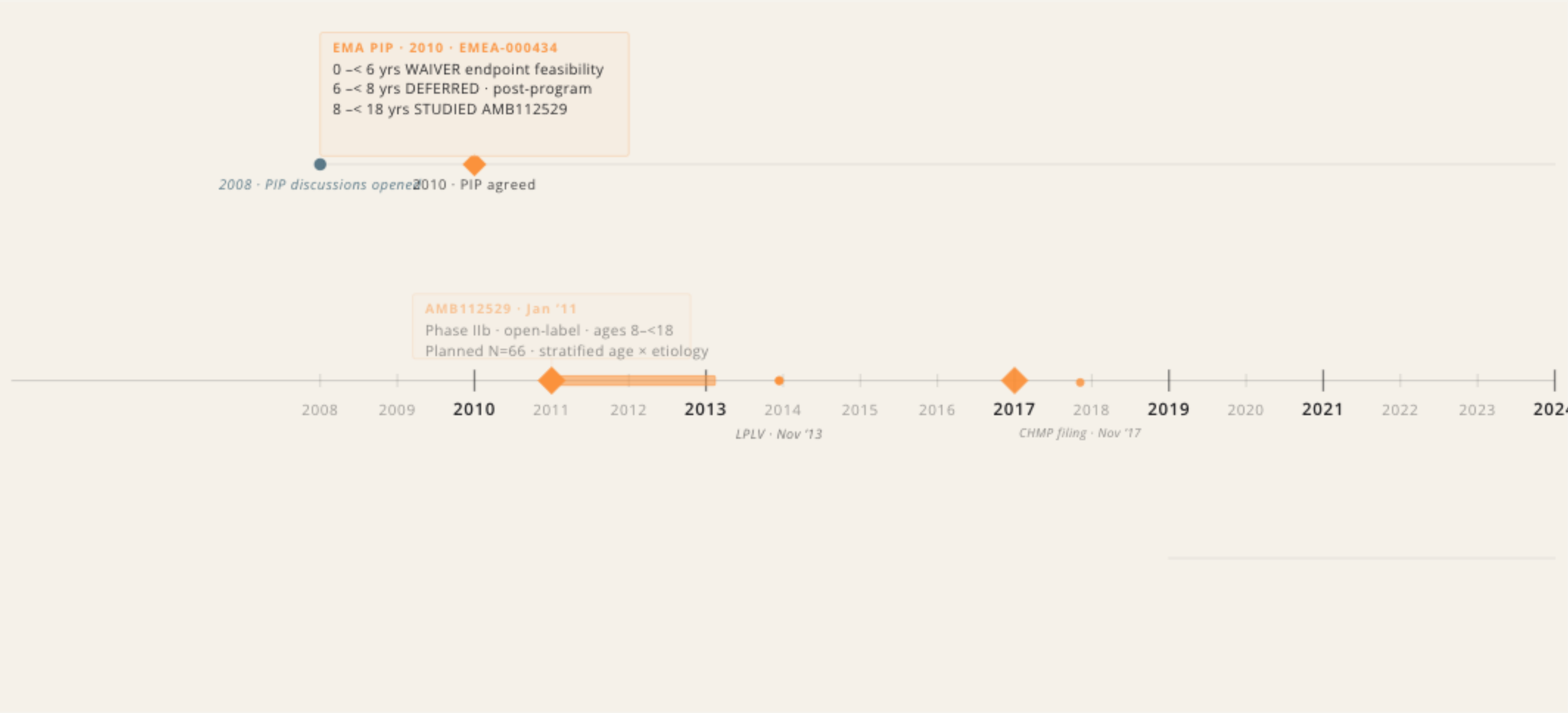
Adult approval to pediatric label · *what the program actually did.*

Adult lifecycle settled by 2010. Pediatric arc: PIP 2010 → trial → 2013 hold → 2017 mechanism → 2019 termination → 2021 dual approval. Framework absorbed the disruption.



↑ ADULT LIFECYCLE

↓ PEDIATRIC PROGRAM · AMB112529



WEIGHT-BANDED DOSING · AMB112529		
	Low arm start / wk2+	High arm start / wk2+
20 –< 35 kg	2.5 / 2.5	2.5 / 5
35 –< 50 kg	5 / 5	5 / 7.5
≥ 50 kg	5 / 5	5 / 10
mg QD · low arm fixed for 24 wks · high arm: 2-wk run-in then up-titration		

TARGET VS ACTUAL · AMB112529	
Planned enrollment	66
Randomized at hold	41
PK-evaluable final	39
Age (eligible)	8 to <18
Age (actual)	8 to 16
Weight floor	20 kg
Weight range	20.1 – 77.0 kg
Treatment duration	24 weeks

From 6MWD primaries to composites — *the field shifted in 2013.*

YEAR	DRUG	PRIMARY ENDPOINT
1995	Epoprostenol	6MWD
2001	Bosentan	6MWD
2002	Treprostinil IV	6MWD
2003	Tadalafil	6MWD
2004	Iloprost	• Composite (first)
2005	Sildenafil	6MWD
2007	Ambrisentan THIS CASE	6MWD
2013	Macitentan	• Composite (SERAPHIN)
2013	Riociguat	6MWD (PATENT-1)
2015	Selexipag	• Composite (GRIPHON)
2024	Sotatercept	6MWD (STELLAR)

The 2013 shift was driven by **Gabler et al. (Circulation 2012)** meta-analysis showing 6MWD alone was insufficient as a surrogate for clinical outcomes. Pediatric programs cannot practically use composite endpoints (too few events, too long to accrue) → **exposure-matching becomes the practical anchor** for extrapolation.

A defensible pediatric dose survived *four constraints*.

Adult evidence supplied the anchor; pediatric evidence confirmed the bridge.

01 · ADULT ANCHOR

Mature adult dose

380 adults · 3,126 PK observations · 6 placebo-controlled studies

Exposure-response already characterized; pediatric work did not invent the therapeutic window.

02 · PEDIATRIC REALITY

Trial cannot carry it

39 PK-evaluable patients · open-label · PK + safety primary

Rare disease, ethical placebo limits, endpoint weakness, and sparse blood sampling define the study.

03 · FIELD WARNING

Empirical dose escalation became unsafe

STARTS-2 · high-vs-low mortality HR 3.95

Sildenafil did not create the architecture; it made dose-selection conservatism unavoidable.

04 · PROGRAM HEADWINDS

Three constraints hit the program

Juvenile rat hold · formal 2019 termination · split US / EU rights

The package had to survive safety scrutiny, incomplete enrollment, and a filing geography problem.

05 · DECISION EVIDENCE

Exposure matching carried the label

Low dose -3% · high dose +0.3% vs adult AUC

EMA and PMDA accepted the totality: adult efficacy, pediatric PK, safety, and LTE follow-up.

Memory hook: **Anchor · Headwinds · Label**. The model did not replace clinical pharmacology; it made the clinical pharmacology decision defensible when the usual trial could not.

— BACKUP TYPE 2

Methodology

The model-informed scaffolding that turned no-trial-feasible into label-defendable: dosing scheme, allometric extrapolation, endpoint defense, Bayesian borrowing, and the PopPK fit + covariate work that anchored exposure-matching.

HOSTILE PROBE *"Walk me through the PopPK model — parameters, diagnostics, and how you justified the 3% AUC match."*

Three weight bands × two dose levels — *six regimens, all matching adult AUC.*

	LOW DOSE ARM		HIGH DOSE ARM	
Weight band	Start	Wk 2+	Start	Wk 2+
≥20 – <35 kg	2.5 mg	2.5 mg	2.5 mg	5 mg
≥35 – <50 kg	5 mg	5 mg	5 mg	7.5 mg
≥50 kg	5 mg	5 mg	5 mg	10 mg

ADULT COMPARATOR

Fixed dose: 5 mg or 10 mg QD · no weight banding

AUC PARITY

Pediatric AUC matches adult within **-3%** (low) and **+0.3%** (high)

Fixed exponents — *mechanistic, not a shortcut.*

ANDERSON-HOLFORD FRAMEWORK

Parameter	Scaling	Rationale
CL	WT^0.75	Metabolic rate (Kleiber's law)
V	WT^1.0	Volume of distribution (body composition)
t½	WT^0.25	Derived: V / CL

1 MECHANISTIC BASIS

Kleiber's law — basal metabolic rate scales as WT^0.75 across species and age

2 SPARSE PEDIATRIC DATA

N = 39 PK-evaluable; estimating free exponents risks over-fitting

3 ESTIMATION ATTEMPTED

Free-exponent model run as sensitivity — OFV drop within noise ($\Delta < 3.84$)

4 CONVENTION

Fixed allometry pre-empts over-parameterization; standard per FDA/EMA pediatric guidance

6MWD isn't a defensible primary efficacy endpoint in pediatric PAH.

01

Developmental

Children <7–8 years cannot perform the test reliably — coordination, comprehension, and stamina vary widely.

02

Motivational

Effort-dependent endpoint; children are inconsistent in giving "best effort" across visits.

03

Growth-confounded

6–12 month trials overlap with growth spurts — baseline distance shifts independent of treatment.

04

Etiologic heterogeneity

IPAH vs CHD-associated PAH have different baseline 6MWD; pooling obscures treatment signal.

05

Ceiling / floor

Mild adolescents walk near healthy norms (ceiling); severe toddlers cannot walk at all (floor).

No pediatric PAH trial has hit a 6MWD primary at $\alpha = 0.05$.

Bayesian borrowing makes the extrapolation argument *formal, not narrative.*

AMB112529 was descriptive by design — no statistical testing. A future program would benefit from prespecified Bayesian borrowing.

HIERARCHICAL / POWER PRIORS

Adult treatment effect becomes an informative prior on pediatric effect, borrowing weight calibrated to adult–pediatric similarity.

DIA Bayesian Scientific Working Group formalized for pediatric extrapolation.

PROPENSITY-SCORE-MATCHED BAYESIAN BORROWING

Only the most clinically relevant adult subjects — matched on baseline severity, etiology, prior therapy — contribute to the prior.

Addresses "adults aren't kids" objection directly.

PHARMACOMETRICS-ENHANCED BAYESIAN BORROWING (PEBB)

Adult PK/PD model builds the prior, updated with pediatric data. DINAMO pediatric T2D trial precedent — FDA input on effective sample size and weight of informative prior.

Closest to the architecture our case used, formalized.

The AMB112529 case is the non-Bayesian version of the same extrapolation argument. PEBB would make the borrowing **explicit and quantifiable**.

Final integrated adult-pediatric PopPK. *Table S3 — side-by-side.*

Parameter estimates, %RSE, IIV — pediatric vs adult final models.

TABLE S3 · PEDIATRIC VS ADULT FINAL MODEL

PARAMETER	PED EST.	PED %RSE	PED IIV (CV%)	PED SHRINKAGE	ADULT EST.	ADULT IIV (CV%)
CL/F (L/hr)	1.17	6.33	21.8	19.0%	1.56	32.1
Vc/F (L)	12.3	16.1	94.1	16.5%	9.52	20.5
Q/F (L/hr)	0.457	21.1	42.5	29.6%	0.928	—
Vp/F (L)	81.3	24.5	33.2	86.1%	8.51	25.3
Ka (1/hr)	2.46	25.7	133	25.9%	1.72	82.6
t_lag (hr)	0.525 FIXED	14.7	31.2	26.4%	0.423	0.036
CL/F ~ WT exponent	0.75 FIXED	—	—	—	0 FIXED	—
Vc/F ~ WT exponent	1.0 FIXED	—	—	—	0.740	—
Prop. residual (CV%)	0.210	14.2	45.8	—	5.89 (RSE 27.6)	—

COVARIANCE

$\eta_{CL} \leftrightarrow \eta_{VC}$: **R = -0.62** — negative correlation as expected: higher clearance pairs with lower central volume. Adult residual error also carried an IIV term on ϵ : **CV 18.2% (RSE 29.9)**.

STRUCTURAL MODEL · ESTIMATION

Two-compartment, 1st-order absorption with **t_{lag}**. Allometric scaling **fixed: 0.75** (CL/F, Q/F), **1.0** (Vc/F, Vp/F) — estimating exponents was attempted but did not improve fit. 70-kg reference. NONMEM 7.4.1 · IS-EM / IMPMAP.

WHY VP/F DIFFERS — AND WHY IT DOESN'T MATTER

Pediatric Vp/F **81.3 L** vs adult **8.51 L**: driven by (i) sample size (ped 211 obs / 39 subj vs adult 3,126 obs / 380 subj) and (ii) sparse vs intensive sampling. Model-based sensitivity analysis confirmed **no clinically relevant difference in systemic exposure** — all pediatric AUC_{SS} within adult model-predicted range.

Model adequacy confirmed — *across every diagnostic.*

Exposure inference — not parameter inference — drove the regulatory conclusion. Both were robust.

PCVPC · 500 REPLICATES
(FIGURE 2)

Observed **median and 5th/95th percentiles** fall within simulated prediction intervals across the full dosing window. **No systematic bias.** Model reproduces the central tendency and spread of the pediatric concentration data.

GOODNESS-OF-FIT · 4 PANELS
(FIGURE S2)

OBS vs IPRED on line of identity	OBS vs PRED scattered — no bias
CWRES vs TIME symmetric ~0 · no trend	CWRES vs PRED symmetric · no trend

VP/F SENSITIVITY · DATA S1

Vp/F fixed at **8.51 · 81.3 · 180 L** across 5 body weights; both dose levels. Simulated AUC_{ss} and C_{max,ss} **did not meaningfully differ** — Vp/F is near-inestimable from sparse data, but exposure predictions are robust to it.

20 / 27.5 / 42.5 / 50 / 70 kg tested at all three Vp/F values · all ≈ equivalent in AUC_{ss} & C_{max,ss}.

ALLOMETRIC PERTURBATION · COVARIATE STABILITY

±10% perturbations on the fixed allometric exponents (**0.75, 1.0**) produced **no material change** in derived AUC_{ss} or C_{max} across the pediatric weight range. Parameter estimates **stable** across with/without each tested covariate.

SPARSE-SAMPLING SCHEDULE
(AMB112529)

- WEEK 4 · trough
- WEEK 8 · 0.5–4 h post-dose
- WEEK 12 · trough
- WEEK 16 · 0.5–4 h post-dose
- WEEK 20 · 4–22 h post-dose
- WEEK 24 · 24 h (trough)

"TWICE-DAILY" FIGURE 2
ARTIFACT

The apparent two-peak appearance in Figure 2 is a **sampling-schedule visualization artifact** — one sample/visit plotted by planned post-dose time across six visits. **No evidence** of delayed absorption or enterohepatic recirculation; under actual QD dosing the PK profile is monotonic.

Pediatric AUC_{ss} matched adult target *within ~3% at weight-based doses.*

Table S5 · model-derived steady-state exposures (geometric mean, 95% CI).

TABLE S5 · AUC _{ss} (MG·HR/ML) · C _{max,ss} (NG/ML)				
GROUP	DOSE	WEIGHT BAND	AUC _{ss} (95% CI)	C _{max,ss} (95% CI)
Adult	5 mg (low)	—	4.98 (4.68–5.29)	469 (447–493)
Adult	10 mg (high)	—	9.12 (8.30–10.0)	830 (757–909)
Pediatric	Low (all)	all bands	4.82 (4.14–5.61)	519 (458–589)
Pediatric	High (all)	all bands	9.15 (8.41–9.96)	981 (894–1080)
Pediatric	Low	≥20 to <35 kg	4.03 (3.48–4.68)	478 (401–568)
Pediatric	Low	≥35 to <50 kg	6.42 (5.09–8.11)	516 (408–651)
Pediatric	Low	≥50 kg	3.87 (2.73–5.48)	624 (369–1060)
Pediatric	High	≥20 to <35 kg	8.73 (7.75–9.83)	953 (850–1070)
Pediatric	High	≥35 to <50 kg	8.70 (7.62–9.92)	1100 (679–1790)
Pediatric	High	≥50 kg	10.2 (8.04–12.8)	948 (791–1140)

Pediatric = aged 8 to <18 years. 95% CI based on percentiles of the sampling distribution.

KEY FINDING

AUC_{ss} bridging confirmed: pediatric low-dose **~3% lower** than adult 5 mg QD; pediatric high-dose **essentially identical** to adult 10 mg QD. Geometric-mean C_{max,ss} was **11% higher** (low dose) and **18% higher** (high dose) vs adults — not considered clinically relevant; no association between predicted AUC_{ss} and incidence of ambrisentan-related AEs (Figure 5).

OUTLIER · ≥35 TO <50 KG LOW-DOSE

Geometric-mean AUC_{ss} **29% higher** than adult 5 mg QD. Explanation: patients in both ≥35–<50 kg and ≥50 kg bands received the same 5 mg dose — relatively higher exposure in the lighter subgroup. **N = 8**; interpret with caution. Individual AUC_{ss} values remained within the adult model-predicted range.

REGULATORY OUTCOME

EMA (Volibris variation X-0061-G): pediatric indication approved ages 8–17 with weight-based dosing. **CHMP positive opinion 22 Jul 2021** (EMA/CHMP/398036/2021); **EC variation Sept 2021** (EMA/CHMP/476747/2021). Independent cohort **Takatsuki et al. N=38** (ages 3–15) confirmed similar PK to adults — external validation.

— BACKUP TYPE 3

Data Cuts

Sub-analyses, longitudinal extensions, and the hemodynamic substudy that confirmed adult-equivalent response in the pediatric population — the second-look data the 6MWD primary doesn't surface.

HOSTILE PROBE *"What does the long-term extension look like? Any DDI signals with PDE-5i co-administration?"*

3.5 years median exposure — *the label held.*

LTE ENROLLMENT 38 / 41 93% of trial completers	COMPLETED (AGED OUT AT 18) 21 / 38	DEATHS 7 / 38 18% · PAH-related, none attributed to ambrisentan	MEDIAN EXPOSURE ~3.5 yr	6MWD IMPROVEMENT ~17% mean, n = 29
WHO FC MAINTAINED OR IMPROVED 100% over full LTE duration	HEPATOTOXICITY None novel distinguishes from bosentan	ANEMIA No novel beyond ERA class expectations	PUBERTAL DEVELOPMENT Collected through age-20 follow-ups	

Ambrisentan doesn't induce CYP3A4 — *no PDE-5i interaction.*

ERA	CYP3A4 INDUCTION	SILDENAFIL AUC	TADALAFIL AUC
Bosentan	Strong (3A4 + 2C9)	↓ 63–73%	↓ ~42%
Macitentan	Minimal	Negligible	Negligible
Ambrisentan	None	None	None

AMB112529 CONTEXT

44% of patients on PDE-5i background therapy at enrollment.

PDE-5i not formally tested as a covariate in the PopPK model — mechanistic defense: **ambrisentan does not induce CYP3A4**, so no pharmacokinetic interaction is expected.

N=5 paired low-dose — *small but directionally consistent with adult ERA effect.*

7

patients with baseline RHC

5

paired (all low-dose)

0

high-dose paired

LOW-DOSE WEEK 24 CHANGES (N = 5)			
PARAMETER	BASELINE MEAN [SD]	WEEK 24 Δ [SD]	DIRECTION
Cardiac index	3.42 [0.50] L/min/m ²	+0.94 [0.66]	↑ Improved
Mean PAP	47.40 [18.96] mmHg	-2.20 [6.06]	↑ Improved
PVR	11.88 [8.30] WU	-3.46 [1.90]	↑ Improved

— BACKUP TYPE 4

Risk Mitigation

The risk surface the program had to manage: a phase-3 efficacy signal that stopped a sister ERA, a preclinical reproductive-tox finding that constrained dosing, and the FDA-side submission gap that opened the divergent verdict landscape.

HOSTILE PROBE *"Wasn't there a safety signal in STARTS that should have stopped the whole class?"*

The safety signal that shaped the discipline — *and its resolution.*

STARTS-1 · BARST 2012

N = 235 · sildenafil TID × 16 wks in pediatric PAH

Primary endpoint: peak VO₂ change · p = 0.056 (missed)

Weight-banded TID dosing: low / medium / high arms

6MWD improved in older subgroup; younger subgroup showed no signal

STARTS-2 · BARST 2014 (LTE)

Mortality signal: HR 3.95 high- vs low-dose

37 deaths during long-term extension

Confounders flagged: sicker patients up-titrated to high dose; disease progression, not drug causation

FDA: sildenafil label added pediatric warning (2012), later revised

RESOLUTION · AFFILIATE 2024

Hoeper et al. Circulation 2024 · N = 385

80 mg TID non-inferior to 5 mg TID (primary endpoint)

FDA sildenafil pediatric approval · 2023 (post-AFFILIATE data)

Juvenile rat finding — *mechanism-specific, age-window-limited, margin-defended.*

THE FINDING

- 3–8% brain weight decrease at 20 mg/kg/day
- Postnatal day 7 (PND7) · dose-dependent · window-specific

MECHANISM

- Sustained hypoxemia from laryngeal-anatomy interaction
- Edema × soft PND7 larynx → airway narrows → intermittent hypoxia
- Reversible ~10 days off-dose

HUMAN RELEVANCE ASSESSMENT

- At-risk window: 0–3 years
- Trial population: ages 8+ · past critical window
- Exposure margin: 1.8–7× human AUC at 10 mg

CONCLUSION · 0–<3 YR WAIVER; 8–<18 YR STUDIED

- EMA PIP: formal waiver below 6 yrs (endpoint feasibility + safety margin)

LONG-TERM EXTENSION · 3.5-YEAR SAFETY

No novel safety signals · LTE 3.5 years

7 / 38 deaths · none attributed to drug (disease progression)

6MWD improvement maintained: +17% from baseline

FDA never received the package — *a commercial structure, not a scientific rejection.*

GSK / VOLIBRIS → EMA + PMDA

- 2008 EMA approves Volibris (adult)
- 2010 EMA PIP agreed · EMEA-000434
- 2011 AMB112529 first patient enrolled
- 2021 EMA pediatric approval (Volibris)
- 2021 PMDA pediatric approval (Japan)

GILEAD / LETAIRIS → FDA

- 2007 FDA approves Letairis (adult)
- 2022 Letairis goes generic (US)
 - No pediatric sNDA filed to FDA
 - No PREA / PSP requirement triggered

FDA label states: "**Safety and effectiveness in pediatric patients have not been established**" — *never evaluated, not rejected.*

— BACKUP TYPE 5

Regulatory Precedent

The regulatory frameworks the AMB112529 program maps onto — pediatric extrapolation guidance, exposure-based dose justification, PIP architecture, and the EMA addendum that updates the post-ICH-E11A landscape.

HOSTILE PROBE *"Is this just AMB112529 or does it map onto ICH E11A and the new EMA pediatric PAH addendum?"*

The case prefigured ICH E11A — *exposure matching codified.*

HIGH SIMILARITY Exposure matching alone may suffice	PARTIAL SIMILARITY Exposure matching + limited efficacy / PD	LIMITED SIMILARITY Full pediatric trial may be required
---	--	---

WHERE AMBRISENTAN SITS

HIGH for disease similarity, pharmacology similarity, and response similarity — exposure matching was defensible as the primary pediatric evidence.

Garnett-Florian: PVR → 6MWD bridging — *FDA's parallel architecture.*

STEP 1 · ADULT PVR-6MWD REGRESSION

12 trials · 2,028 patients · 9 drugs · 5 classes

Slope: -0.055 m per dyne·sec/cm⁵

~49% of treatment effect explained by PVR change

STEP 2 · PEDIATRIC PVR DATA

BREATHE-3: N = 19

Δ PVR ≈ -389 dyne·sec/cm⁵ (bosentan)

Largest pediatric hemodynamic dataset available

STEP 3 · STOCHASTIC SIMULATION

500 simulated datasets

Based on Δ PVR mapped through adult regression

Monte Carlo propagation of uncertainty

STEP 4 · PREDICTED PEDIATRIC 6MWD

+14 m (95% CI: 3–31 m)

CI excludes zero → bridging accepted

Directionally consistent with adult ERA effect

Three age bands, three regulatory decisions — *waiver, deferral, study.*

AGES 0–6 · WAIVER GRANTED

PDCO 2010

Endpoint feasibility

6MWD / WHO FC not reliably measurable below age 6–7.

Disease paradigm

Sub-6 PAH includes PPHN, uncorrected CHD, developmental lung disease — adult ERA pharmacology generalizes less cleanly.

"Waiver predates the juvenile rat finding by 3 years."

AGES 6–<8 · DEFERRED

not waived

Regulatory commitment

Study required after 8–<18 program completes.

Practical execution unlikely

Post-2013 juvenile rat finding — at-risk window 0–3 but safety margin uncomfortable for expansion.

"Deferral technically still on books."

AGES 8–<18 · STUDIED

AMB112529

41 enrolled, 39 PK-evaluable

EMA pediatric label April 2021.

PMDA March 23, 2021

Japanese pediatric approval followed EMA.

"This is what was studied, approved, and presented."

20 kg weight floor excluded most children under 8 regardless of age eligibility.

The EMA addendum now *formalizes* what the 2023 ambrisentan approach demonstrated.

Draft CHMP/60723/2026 · adopted 19 March 2026 · methodology published 2023 → framework codified 2026.

ADDENDUM · §4.1.1.1.1 · VERBATIM EXCERPTS

What the addendum actually says.

"Extrapolation of evidence on pharmacokinetics (PK), efficacy and safety, generated in adult patients, is an **important element** of the development programme in paediatric patients with PAH."

CHMP/60723/2026 · §4.1.1.1.1

"Matching exposure that falls into the exposure range of adults or combination of **exposure matching with PK/PD** is an important element contributing to extrapolation based on **exposure-response (E-R)**."

CHMP/60723/2026 · §4.1.1.1.1

ICH E11A

Pediatric extrapolation framework. Defines extrapolation concept and plan.

Adopted Jan 2025

ICH M15

Model-informed drug development. MIDD methodology standards for regulatory submissions.

Current draft / in-force

WHERE BRIDGING WORK IS STILL NEEDED

The open problems the addendum flags.

NOT YET VALIDATED FOR EXTRAPOLATION

- 01 NT-proBNP** — not yet validated as a bridging biomarker for pediatric PAH extrapolation. Adult-to-pediatric mapping remains an open regulatory question.
- 02 Echocardiography-derived endpoints** — not yet validated for extrapolation. Reproducibility and pediatric-specific reference ranges are needed.
- 03 Pediatric-specific risk scores** — adult REVEAL-style composite scores have not been validated in pediatric populations; pediatric-specific scoring is an acknowledged gap.

WHERE THIS WORK GOES NEXT

These are exactly the kinds of questions model-informed frameworks — PopPK/PD, exposure-response, biomarker qualification — are built to answer. The open problems **are** the next generation of pediatric extrapolation work.

THE HONEST CLAIM

Our 2023 methodology *maps onto* what the 2026 addendum now formally recommends. (not "influenced" · not "shaped" · not "contributed to" — maps onto.)

— BACKUP · CASE STUDY 02 · IVOSIDENIB

Ivosidenib

Defense library · pull on demand



— BACKUP TYPE 1

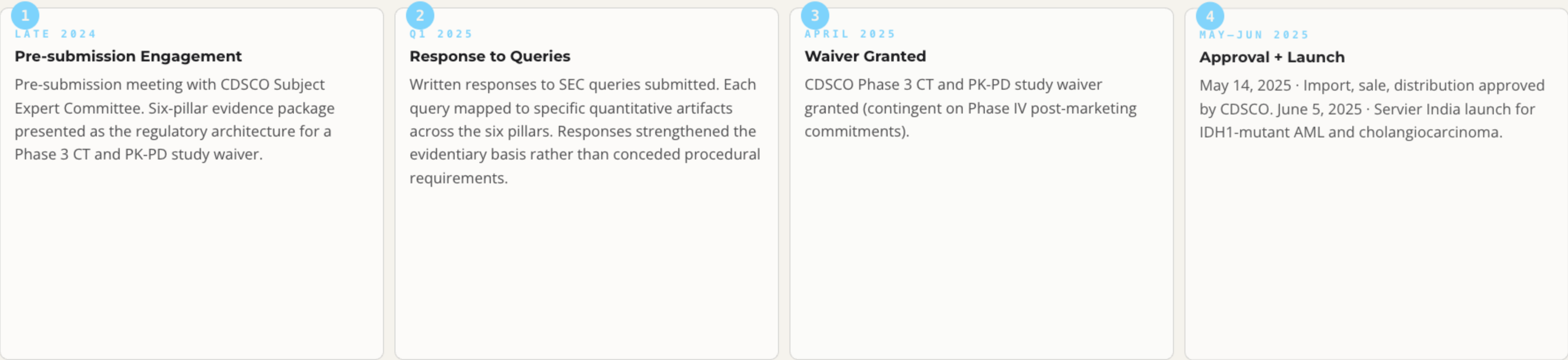
Historical Context

The CDSCO engagement arc — pre-submission through approval and launch. The temporal frame for the regulatory work.

HOSTILE PROBE *"Walk me through your engagement timeline with CDSCO — when did you start, what did each step accomplish?"*

From pre-submission to patient access — *the regulatory path.*

Late 2024 to June 2025 · waiver granted contingent on Phase IV post-marketing commitments.



WHAT THIS TIMELINE ENABLED

\$10M+

in bridging study costs avoided

18–24 months

of Indian patient-access acceleration

Reusable

six-pillar framework adopted as Servier template

Dates from Servier India public press releases and CDSCO public approval record. Specific SEC meeting dates and internal regulatory deliberations are out of public scope. "\$10M+ saved" and "18–24 months" are industry-typical estimates, not Servier-documented.

Methodology

The quantitative architecture that made the waiver defensible — six convergent evidence pillars and the Phase 1 dose rationale that anchored the global label.

Tibsovo India — *the full regulatory architecture.*

Each pillar an independent evidence stream. Convergence across all six is what made the waiver defensible.

1 Global PK Equivalence

- Cross-region PopPK analyses from pivotal studies
- Exposure comparisons across approved populations
- No evidence of region-specific PK shifts

2 IDH1 R132 Mutation Biology

- Acquired, tumor-localized somatic mutation
- ~99% at R132 hotspot (exon 4)
- Mutation biology independent of ethnicity (Tsilimigras 2024 · Suryavanshi 2025)

3 PopPK Race Insensitivity

- Jiang et al. CTS 2021 · race NOT correlated with CL/F (n = 253)
- Yue et al. EJCP 2024 · Chinese-specific PK confirms safe/effective dosing
- No race covariate retained in either analysis

4 Metabolism + DDI

- CYP3A4 primary metabolic pathway
- UGT1A1 polymorphism analysis across populations
- Indian-population DME allele frequency review

5 Exposure–Response Consistency

- E-safety and E-efficacy profiles from pivotal studies
- No race covariate in E–R models
- Consistent exposure–benefit relationship across regions

6 Regulatory Precedent

- 42+ international approvals at time of CDSCO submission
- FDA, EMA, PMDA label alignment
- Rare-disease framework precedents

500 mg QD established across two Phase 1 studies — *AUC plateau, MTD not reached.*

AG120-C-001 (AML) + AG120-C-002 (CCA) · the same dose anchored both indications.

STUDY DESIGN — AG120-C-001 (AML)

Phase 1, multicenter, open-label, dose escalation + expansion

5 dose cohorts: 100 mg BID, 300 mg QD, 500 mg QD, 800 mg QD, 1200 mg QD

Dose escalation n = 67

Dose expansion n = 173

PK AT DOSE

COHORT	AUC _{0–8} (H · NG / ML)	C _{MAX} (NG / ML)
100 mg BID	22,761	2,743
300 mg QD	52,063	6,633
500 mg QD ← APPROVED	56,212	6,710
800 mg QD	59,275	7,280
1200 mg QD	96,865	11,810

STUDY DESIGN — AG120-C-002 (CCA)

Phase 1, multicenter, open-label, dose escalation + expansion

6 dose cohorts (including 400 mg QD arm specific to CCA)

Dose escalation n = 24

Dose expansion n = 49

PK AT DOSE

COHORT	AUC _{0–8} (H · NG / ML)	C _{MAX} (NG / ML)
100 mg BID	20,818	3,220
300 mg QD	19,840	3,157
400 mg QD	28,862	4,880
500 mg QD ← APPROVED	28,877	4,547
800 mg QD	26,775	4,709
1200 mg QD	57,666	9,310

DOSE RATIONALE

Exposure increased less proportionally with dose above 500 mg QD in both indications.

MTD was not reached in either study. The 500 mg QD dose was selected based on the combination of AUC/C_{max} plateau behavior, favorable safety profile, and target engagement evidence. Both studies support 500 mg QD as the optimal clinical dose.

— BACKUP TYPE 3

Data Cuts

Population-specific evidence — IDH1 prevalence and DME polymorphism frequencies across ethnic groups. The subgroup data that defended Pillar 3 (PopPK race insensitivity).

IDH1 prevalence and DME polymorphisms — *the ethnic-evidence base.*

Cross-regional prevalence + Indian-specific cohort + UGT1A1/CYP3A4 frequencies — defends Pillar 3.

CROSS-REGIONAL IDH1 IN ICCA · TSILIMIGRAS ASO 2024 · N = 1,068

POPULATION	IDH1 MUTATION PREVALENCE
White	20.8%
Black	5.6%

INDIAN-SPECIFIC · SURYAVANSHI 2025 · N = 147

Indian cholangiocarcinoma cohort: IDH1 prevalence ~9.5%. Supports unmet need characterization for the Indian patient population.

INTERPRETATION FOR IVOSIDENIB

Ivosidenib is primarily CYP3A4-metabolized. DME polymorphism frequencies differ modestly between populations; the clinical impact on ivosidenib disposition is bounded. **No population-specific dose adjustment is indicated by the evidence** — confirmed by PopPK analyses on Jiang 2021 and Yue 2024 cohorts.